

Complete Growing-Ratio Proof Supplement

Original root forms, monic norms, arithmetic transfer, and shifted action

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This supplement gives the complete new growing-ratio proofs and their backward receiving arguments. It preserves the original root polynomial, Gamma mass, factorial norms, all degree and parity shifts, and every finite constant and domain. The foundational equilibrium, closed-coefficient, and Binet calculations are reproduced in full in the appendices. The complete accompanying cumulative proof edition supplies all unchanged earlier arithmetic, cyclic, kernel, boundary, and signed-return constructions referred to by their original equation tags. Thus the two documents together provide the complete source closure without rewriting the inherited arguments.

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1 The complete root-form comparison at growing degree ratios

Scope and audited conclusion. This section proves equations (2)–(13) of the received submission, including all their finite constants, on the original stipulated complete-order packet. The verdict on that scope is *proved as written*, with the displayed guards retained. The accepted EIQ equilibrium and ECL evaluations are stated below in their exact original domain and normalization; the new full-form argument is proved in full here. No assertion about arithmetic monic minima, allowance asymptotics, shifted windows, an intrinsic mixed allocation, or a period evaluation is made here. In particular no original mass, period, root, or multiplicity is altered.

1.1 Original objects and exact dependency statements

Fix $0 < \delta < 1/2$, $\gamma > 2$, and the positive integer complete order m_{pkt} . Retain the original integer subsequence $k = 4l + 1 \geq 5$ and put

$$e_k = 1 + k(m_{\text{pkt}} - 1), \quad q = e_k(k + 1)^2, \quad c = k/2, \quad R = R_k = k\sqrt{\delta^2 + \gamma^2}.$$

The root polynomial and its exact real-coordinate image are

$$\chi(S) = \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^{e_k},$$

$$Q(y) = i^{-q} \chi(c + iy) = \prod_{a,b=0}^k (y - \zeta_{ab})^{e_k}, \quad \zeta_{ab} = (2b - k)\gamma - i(2a - k)\delta.$$

The symbol m_{pkt} here denotes exactly the received packet order m ; the subscript merely distinguishes it from the degree of a parity block. All q roots, with their complete multiplicities, occur in opposite and conjugate pairs. They satisfy $|\zeta_{ab}| \leq R$.

The measure remains

$$d\sigma(y) = s(y) dy, \quad s(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}.$$

Its accepted original density bounds are, on all of $\mathbb{R}$,

$$c_{\Gamma} e^{-\pi|y|/2} (1 + |y|)^{-1/2} \leq s(y) \leq C_{\Gamma} e^{-\pi|y|/2} (1 + |y|)^{-1/2}, \quad c_{\Gamma} = \sqrt{2}e^{-7/6}, \quad C_{\Gamma} = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{GRF1})$$

These are AMT5/TW4, restated in RGC2; no new constants are substituted. For completeness the mass follows directly from the Gamma integral. Writing $f_a(x) = e^{ax-e^x}$ gives $\Gamma(a + it) = \int_{\mathbb{R}} f_a(x) e^{itx} dx$ for $a > 0$. Plancherel and $z = e^x$ therefore give

$$\int_{\mathbb{R}} |\Gamma(a + it)|^2 \frac{dt}{2\pi} = \int_{\mathbb{R}} e^{2ax-2e^x} dx = \frac{\Gamma(2a)}{2^{2a}}.$$

Setting $a = 1/4$ and $y = 2t$ proves $\sigma(\mathbb{R}) = 2\Gamma(1/2)/\sqrt{2} = \sqrt{2\pi}$.

Here is the precise accepted equilibrium theorem used. For every $\alpha > 0$, $\beta > 0$, let $V(x) = \beta\sqrt{x} - \alpha \log x$ on $(0, \infty)$. The EIQ probability measure has support $[u^2, v^2]$, $0 < u < v$, where

$$uK(\kappa) = \frac{\alpha\pi}{\beta}, \quad vE(\kappa) = \frac{(\alpha + 2)\pi}{\beta}, \quad \kappa^2 = 1 - u^2/v^2, \quad (\text{GRF2})$$

and

$$K(\kappa) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - \kappa^2 \sin^2 \theta}}, \quad E(\kappa) = \int_0^{\pi/2} \sqrt{1 - \kappa^2 \sin^2 \theta} d\theta.$$

The density, extended by zero off this interval, is

$$\rho(x) = \frac{\sqrt{(v^2 - x)(x - u^2)}}{2\pi} h(x), \quad h(x) = \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x + s)\sqrt{(u^2 + s)(v^2 + s)}} ds > 0. \quad (\text{GRF3})$$

It is bounded and continuous, has total mass one, and its logarithmic potential $P^{\text{eq}}(z) = \int_{u^2}^{v^2} \log |z - x| \rho(x) dx$ is continuous on $\mathbb{C}$. With the original Robin constant λ ,

$$V(x) - 2P^{\text{eq}}(x) = \lambda \quad (u^2 \leq x \leq v^2), \quad V(x) - 2P^{\text{eq}}(x) > \lambda \quad (0 < x < u^2 \text{ or } x > v^2). \quad (\text{GRF4})$$

These statements are EIQ1–19 on the entire positive parameter quadrant. Writing $w = (u + v)/2$, $z_0 = uv$, and $c_{\text{cap}} = (v^2 - u^2)/4$, the accepted ECL6 and ECL14 give exactly

$$\lambda = 2(\alpha + 2) - 2\alpha \log w - 2 \log c_{\text{cap}}, \quad P^{\text{eq}}(0) = 2(\alpha + 1) \log w - \alpha \log z_0 - 2. \quad (\text{GRF5})$$

The use of these already proved dependencies does not assume compactness of a moving family of parameters. All estimates below use the actual α, β, u, v .

1.2 The complete root polynomial

Lemma 1. *Put*

$$U_2 = \frac{k(k+2)}{3}, \quad U_4 = \frac{k(k+2)(3k^2+6k-4)}{15},$$

$$D_2 = qU_2(\gamma^2 - \delta^2), \quad M_4 = q[(\delta^4 + \gamma^4)U_4 + 2\delta^2\gamma^2U_2^2].$$

For real $|y| > R$ one has

$$\log \frac{|Q(y)|^2}{y^{2q}} = -\frac{D_2}{y^2} + E_Q(y), \quad |E_Q(y)| \leq \frac{M_4}{2y^4(1 - R^2/y^2)}. \quad (\text{GRF6})$$

For real $|y| \geq 2R$ one has

$$\log \frac{|Q(y)|^2}{y^{2q}} \leq -\frac{13D_2}{30y^2} < 0. \quad (\text{GRF7})$$

For every real y one has

$$|Q(y)|^2 \leq (y^2 + R^2)^q. \quad (\text{GRF8})$$

Proof. The original odd values of k give the literal positive real factorization

$$Q(y) = \prod_{a,b \in \{1,3,\dots,k\}} [((y - b\gamma)^2 + a^2\delta^2)((y + b\gamma)^2 + a^2\delta^2)]^{e_k} > 0.$$

No root is real, and this also shows that Q is even. Finite differentiation of

$$\sum_{a=0}^k e^{(2a-k)t} = \frac{\sinh((k+1)t)}{\sinh t}$$

at $t = 0$, using the removable value there, gives zero odd moments and

$$\frac{1}{k+1} \sum_{a=0}^k (2a-k)^2 = U_2, \quad \frac{1}{k+1} \sum_{a=0}^k (2a-k)^4 = U_4.$$

For example division of the Taylor series gives the coefficients $U_2/2$ and $U_4/24$ in the series of the displayed ratio divided by $k+1$. Expanding ζ_{ab}^2 and $|\zeta_{ab}|^4 = [(2b-k)^2\gamma^2 + (2a-k)^2\delta^2]^2$ therefore proves

$$\sum_{a,b} e_k \zeta_{ab}^2 = D_2, \quad \sum_{a,b} e_k |\zeta_{ab}|^4 = M_4 \leq qR^4.$$

For $|y| > R$ the series for $\log(1 - \zeta_{ab}/y)$ converges absolutely. Taking twice the real part, opposite-root pairing cancels its odd terms, and conjugate pairing makes its even sums real. Hence

$$\log \frac{|Q(y)|^2}{y^{2q}} = - \sum_{j \geq 1} \frac{\sum_{a,b} e_k \zeta_{ab}^{2j}}{j y^{2j}}.$$

This is the real logarithm of a positive number. Since $\sum e_k |\zeta_{ab}|^{2j} \leq M_4 R^{2j-4}$ for $j \geq 2$ and $1/j \leq 1/2$, summing the geometric majorant proves (GRF6).

The packet restrictions give $\delta^2/\gamma^2 < 1/16$, so

$$\frac{D_2}{qR^2} = \frac{k+2}{3k} \frac{\gamma^2 - \delta^2}{\gamma^2 + \delta^2} \geq \frac{5}{17} > 0.$$

At $|y| \geq 2R$, the error divided by D_2/y^2 is at most

$$\frac{qR^2}{D_2} \frac{R^2/y^2}{2(1 - R^2/y^2)} \leq \frac{17}{5} \frac{1/4}{2(3/4)} = \frac{17}{30}.$$

Subtracting this from the negative main term proves (GRF7). Finally pair each root with its opposite, with multiplicities. Every factor satisfies $|(y - \zeta)(y + \zeta)| = |y^2 - \zeta^2| \leq y^2 + R^2$. There are exactly $q/2$ such factors in Q , and squaring their product proves (GRF8). $\square$

1.3 The exact parity coordinate and moving evaluation bound

Let r be an integer, set $m_\varepsilon = \lfloor (r - \varepsilon)/2 \rfloor$ for $\varepsilon \in \{0, 1\}$, and impose $m_0, m_1 \geq 1$. This guard is equivalent to $r \geq 3$ and is retained throughout the finite theorem. Every complex polynomial p of degree at most r has the unique decomposition

$$p(y) = P_0(y^2/q^2) + (y/q)P_1(y^2/q^2), \quad \deg P_\varepsilon \leq m_\varepsilon.$$

Indeed, if $p(y) = \sum_{j=0}^r a_j y^j$, then $P_\varepsilon(x) = \sum_{j \geq 0} a_{2j+\varepsilon} q^{2j+\varepsilon} x^j$, with only indices $2j + \varepsilon \leq r$ included. This gives both the map and its inverse on coefficient spaces. Define, for $y > 0$,

$$\mathfrak{r}(y) = \frac{s(y)e^{\pi y/2} \sqrt{y}}{\sqrt{2}}.$$

For each block let

$$\alpha_\varepsilon = \frac{q + \varepsilon - 3/4}{m_\varepsilon} > 0, \quad \beta_\varepsilon = \frac{\pi q}{2m_\varepsilon} > 0, \quad V_\varepsilon(x) = \beta_\varepsilon \sqrt{x} - \alpha_\varepsilon \log x, \quad (\text{GRF9})$$

and

$$\|P\|_\varepsilon^2 = \int_0^\infty |P(x)|^2 e^{-m_\varepsilon V_\varepsilon(x)} \mathfrak{r}(q\sqrt{x}) dx.$$

Denote its actual equilibrium data by $u_\varepsilon, v_\varepsilon, P_\varepsilon^{\text{eq}}, \lambda_\varepsilon$. Since s and $|Q|^2$ are even, the two half-lines cancel the cross term $2 \operatorname{Re}(P_0 \bar{P}_1) y/q$ for both norms. Using $y = q\sqrt{x}$, $dy = q dx/(2\sqrt{x})$ on the positive half-line gives

$$\mathcal{N}_0[p] := \int_{\mathbb{R}} y^{2q} |p(y)|^2 d\sigma(y) = \sqrt{2} q^{2q+1/2} \sum_{\varepsilon=0}^1 \|P_\varepsilon\|_\varepsilon^2. \quad (\text{GRF10})$$

In particular the power of x in the integrand is $q + \varepsilon - 3/4$, exactly as in received equation (5). With $H_Q(x) = q^{-2q} |Q(q\sqrt{x})|^2$, the other norm is exactly

$$\mathcal{N}_\chi[p] := \int_{\mathbb{R}} |Q(y)p(y)|^2 d\sigma(y) = \sqrt{2} q^{2q+1/2} \sum_{\varepsilon=0}^1 \int_0^\infty |P_\varepsilon(x)|^2 H_Q(x) x^{\varepsilon-3/4} e^{-\pi q \sqrt{x}/2} \mathfrak{r}(q\sqrt{x}) dx. \quad (\text{GRF11})$$

All integrals converge by (GRF1).

For each block put

$$\begin{aligned}\Sigma_\varepsilon &= m_\varepsilon(2P_\varepsilon^{\text{eq}}(0) + \lambda_\varepsilon), & b_\varepsilon &= \min \left\{ \frac{u_\varepsilon}{2}, \frac{1}{4}e^{-\Sigma_\varepsilon/(2q)} \right\} > 0, \\ \Lambda_\varepsilon &= \frac{\alpha_\varepsilon}{u_\varepsilon^2} + \frac{\beta_\varepsilon}{2u_\varepsilon}, & r_+ &= \frac{C_\Gamma}{\sqrt{2}}, & r_{-, \varepsilon} &= \frac{c_\Gamma}{\sqrt{2}} \left(\frac{qu_\varepsilon}{1 + qu_\varepsilon} \right)^{1/2} > 0.\end{aligned}\tag{GRF12}$$

The original density bounds imply $\mathfrak{r}(y) \leq r_+$ for all $y > 0$ and $\mathfrak{r}(q\sqrt{x}) \geq r_{-, \varepsilon}$ on the actual support. Indeed the extra factor is $\sqrt{y/(1+y)}$, which increases in y . Also $|V'_\varepsilon(x)| \leq \Lambda_\varepsilon$ on that support.

Adding the two exact expressions in (GRF5) yields, for each parameter pair and its original w, z_0 ,

$$2P^{\text{eq}}(0) + \lambda = 2\alpha \left(1 + \log \frac{w}{z_0} \right) + 2 \log \frac{v+u}{v-u}.\tag{GRF13}$$

Here $4 \log w - 2 \log c_{\text{cap}} = 2 \log((v+u)/(v-u))$ and the constants give $-4 + 2(\alpha + 2) = 2\alpha$. Thus the -2 in each occurrence of $P^{\text{eq}}(0)$ is retained.

Lemma 2. Assume $m = m_\varepsilon \geq 1$ and $m(v_\varepsilon^2 - u_\varepsilon^2) \geq 2$. For every complex polynomial P of degree at most m and every $0 \leq x \leq b_\varepsilon^2$,

$$|P(x)|^2 \leq \frac{32e^{3\Lambda_\varepsilon/2}m^4}{r_{-, \varepsilon}} e^{\Sigma_\varepsilon} \|P\|_\varepsilon^2.\tag{GRF14}$$

Proof. Suppress the block subscripts only within this proof, retaining their actual values. If $P = 0$ the result holds. Otherwise let $M = \max_{[u^2, v^2]} |P(x)|e^{-mV(x)/2} > 0$, attained at x_0 . At least one side of x_0 in this interval has length at least $1/m$, because its full width is at least $2/m$. Take the closed one-sided interval J of length $1/m$ from x_0 . For $x \in J$,

$$|P(x)| \leq M e^{mV(x)/2} \leq e^{\Lambda/2} |P(x_0)|.$$

For any interval J and every complex polynomial of degree at most m ,

$$\|P'\|_{\infty, J} \leq \frac{4m^3}{|J|} \|P\|_{\infty, J}.\tag{GRF15}$$

To prove this, map J affinely to $[-1, 1]$ and expand $P = \sum_{j=0}^m c_j T_j$, where $T_j(\cos \theta) = \cos(j\theta)$. Cosine orthogonality gives $|c_j| \leq 2\|P\|_\infty$ for $j \geq 1$, also for complex c_j . The finite geometric-sum identity for $\sin(j\theta)/\sin \theta$ gives its modulus at most j . Thus $|T'_j| \leq j^2$, including the endpoints by continuity. Since $\sum_{j=1}^m j^2 \leq m^3$ for every integer $m \geq 1$, the derivative bound on $[-1, 1]$ is $2m^3\|P\|_\infty$; the affine derivative factor $2/|J|$ proves (GRF15).

On J , this gives $|P'| \leq 4m^4 e^{\Lambda/2} |P(x_0)|$. Consequently on the segment from x_0 of length $d = (8e^{\Lambda/2}m^4)^{-1} \leq 1/m$, the fundamental theorem of calculus and the triangle inequality imply $|P| \geq |P(x_0)|/2$. On the same segment $e^{-mV(x)} \geq e^{-\Lambda} e^{-mV(x_0)}$. Using the actual lower density r_- on it proves

$$\|P\|_\varepsilon^2 \geq r_- d \frac{|P(x_0)|^2}{4} e^{-\Lambda} e^{-mV(x_0)} = \frac{r_- M^2}{32e^{3\Lambda/2}m^4}.$$

For the passage off the support, continuity on the whole complex plane can be checked directly from the accepted bounded density. If its bound is K_ρ , then for $0 < a < 1$ and every complex z ,

$$\int_{|z-t|<a} -\log |z-t| \rho(t) dt \leq 2K_\rho a(1 - \log a).$$

Indeed the real length of $\{t : |z-t| < s\}$ is at most $2s$; integrating the level sets of the negative logarithm gives the stated bound. It tends uniformly to zero as $a \downarrow 0$. Truncation of this singularity gives continuous potentials converging locally uniformly, while the positive logarithmic part is

continuous on compact sets. Thus P^{eq} is continuous on $\mathbb{C}$. Outside the support its integrands are harmonic and uniformly differentiable on compact subsets, so P^{eq} is harmonic there. Now consider $f(z) = \log |P(z)| - mP^{\text{eq}}(z)$ on $\mathbb{C} \setminus [u^2, v^2]$. The second term is harmonic there and the first is subharmonic, including its value $-\infty$ at zeros. Continuity of P^{eq} and (GRF4) give boundary upper values $f \leq \log M + m\lambda/2$. This inequality also bounds the entire exterior, including infinity: if $n = \deg P \leq m$ and $w = 1/z$, then near $w = 0$

$$f(1/w) = (m - n) \log |w| + \log |w^n P(1/w)| - m \int \log |1 - tw| \rho(t) dt.$$

The last two terms are harmonic near zero because their arguments are nonzero there. For $n = m$ the function extends harmonically at infinity; for $n < m$ it extends subharmonically with value $-\infty$. Thus it is subharmonic on the complement of the interval in the Riemann sphere. Its boundary values have the stated upper bound. The maximum principle applies on this compact closure: an upper value greater than the boundary bound would attain an interior maximum, forcing constancy on the connected exterior, contrary to the boundary bound. Hence

$$|P(z)|^2 \leq M^2 e^{m(2P^{\text{eq}}(z) + \lambda)}.$$

For $0 \leq x < u^2$, every support point t satisfies $0 < t - x \leq t$, so $\log |x - t| \leq \log t$ and $P^{\text{eq}}(x) \leq P^{\text{eq}}(0)$. Since $b \leq u/2$, this applies throughout $[0, b^2]$. Substituting the proved estimate for M^2 yields (GRF14). $\square$

1.4 The entire inner region and the finite full-form theorem

Let $\varrho = R/q$. Impose, for both blocks,

$$m_\varepsilon \geq 1, \quad m_\varepsilon(v_\varepsilon^2 - u_\varepsilon^2) \geq 2, \quad \varrho \leq b_\varepsilon/2. \quad (\text{GRF16})$$

Set

$$\eta_\varepsilon = \frac{32r_+ e^{3\Lambda_\varepsilon/2} m_\varepsilon^4 b_\varepsilon^{2\varepsilon+1/2}}{r_{-, \varepsilon}(\varepsilon + 1/4)} 8^{-q}, \quad \eta = \max(\eta_0, \eta_1). \quad (\text{GRF17})$$

On $[0, b_\varepsilon^2]$, (GRF8) gives $H_Q(x) \leq (x + \varrho^2)^q$. The cutoff definition and the third guard give

$$e^{\Sigma_\varepsilon} (b_\varepsilon^2 + \varrho^2)^q \leq e^{\Sigma_\varepsilon} (5b_\varepsilon^2/4)^q \leq (5/64)^q \leq 8^{-q}.$$

The same estimate holds with x^q instead of $H_Q(x)$. Multiplying (GRF14) by the remaining factors in (GRF11), and using $e^{-\pi q \sqrt{x}/2} \leq 1$ and $\mathfrak{r}(q\sqrt{x}) \leq r_+$, bounds each entire inner block by

$$\frac{32r_+ e^{3\Lambda_\varepsilon/2} m_\varepsilon^4}{r_{-, \varepsilon}} 8^{-q} \|P_\varepsilon\|_\varepsilon^2 \int_0^{b_\varepsilon^2} x^{\varepsilon-3/4} dx = \eta_\varepsilon \|P_\varepsilon\|_\varepsilon^2.$$

The integral is finite for both $\varepsilon = 0, 1$, and its exact value is $b_\varepsilon^{2\varepsilon+1/2}/(\varepsilon + 1/4)$. Thus this estimate includes the endpoint zero and every neighborhood of it.

Define the finite quantity

$$C_\chi = \max_{\varepsilon=0,1} \left\{ \frac{D_2}{q^2 b_\varepsilon^2} + \frac{M_4}{2q^4 b_\varepsilon^4 (1 - R^2/(q^2 b_\varepsilon^2))} \right\}. \quad (\text{GRF18})$$

Its denominators are positive, at least $3/4$ in the parenthesis, by (GRF16).

Theorem 3. *Under (GRF16) and $\eta < 1$, for every complex polynomial p of degree at most r ,*

$$e^{-C_\chi} (1 - \eta) \mathcal{N}_0[p] \leq \mathcal{N}_\chi[p] \leq (1 + \eta) \mathcal{N}_0[p]. \quad (\text{GRF19})$$

Proof. In block ε , the outer interval $x \geq b_\varepsilon^2$ corresponds to $|y| \geq qb_\varepsilon \geq 2R$. By (GRF6) and monotonicity of its displayed majorants there, the density ratio $H_Q(x)/x^q$ is at least e^{-C_χ} . By (GRF7) the same ratio is at most 1. Write $N_\varepsilon = \|P_\varepsilon\|_\varepsilon^2$. The proved homogeneous inner bound implies its homogeneous outer contribution is at least $(1 - \eta_\varepsilon)N_\varepsilon$. Positivity of the root inner contribution then gives the lower bound $e^{-C_\chi}(1 - \eta_\varepsilon)N_\varepsilon$ for the root block. For the upper bound, its outer contribution is at most N_ε and its entire inner contribution is at most $\eta_\varepsilon N_\varepsilon$. Sum the two block inequalities with the common, unchanged factor $\sqrt{2}q^{2q+1/2}$ of (GRF10)–(GRF11). The result is (GRF19). No assumption on the location of polynomial mass and no monic minimization has been used. $\square$

1.5 Proof of the broad growing-ratio range

Theorem 4. *Keep the fixed original packet and let $k = 4l + 1 \rightarrow \infty$. Let $r = r_k$ be an integer, $t = r/q$, and suppose*

$$t \rightarrow \infty, \quad \log t = o(q/R). \quad (\text{GRF20})$$

Then both blocks eventually satisfy every guard in Theorem 3. More precisely,

$$u_\varepsilon \sim \frac{2}{\log t}, \quad v_\varepsilon \sim 2t, \quad b_\varepsilon \sim \frac{u_\varepsilon}{2e} \sim \frac{1}{e \log t}, \quad (\text{GRF21})$$

$$C_\chi = O_h\left(\frac{R^2}{q}(1 + \log t)^2\right) = o(q), \quad \eta = \exp[-q \log 8 + o(q)]. \quad (\text{GRF22})$$

The corresponding original reference endpoint u_t at $(\alpha, \beta) = (2/t, \pi/t)$ has $u_t \sim 2/\log t$, so $b_\varepsilon \sim u_t/(2e)$ and $qb_\varepsilon \sim q/(e \log t)$.

Proof. All limits below hold for each of the two possible block indices, so are uniform over that finite set. Put

$$c_\varepsilon = \varepsilon - 3/4, \quad d_\varepsilon = r - 2m_\varepsilon \in \{\varepsilon, \varepsilon + 1\}, \quad A_\varepsilon = 1 + c_\varepsilon/q, \quad B_\varepsilon = A_\varepsilon + 2m_\varepsilon/q.$$

These are exact identities; in particular $B_\varepsilon = 1 + t + (c_\varepsilon - d_\varepsilon)/q$. The endpoint equations (GRF2) at the actual parameters become

$$u_\varepsilon K(\kappa_\varepsilon) = 2A_\varepsilon, \quad v_\varepsilon E(\kappa_\varepsilon) = 2B_\varepsilon. \quad (\text{GRF23})$$

Here $A_\varepsilon \rightarrow 1$, $B_\varepsilon/t \rightarrow 1$, and $m_\varepsilon \sim qt/2$. Thus $m_\varepsilon \geq 1$ eventually.

We prove the endpoint asymptotics by the defining elliptic integrals, without importing a uniform remainder estimate. Write $s = u/v \in (0, 1)$, $L = \log(1/s)$, and suppress the block index for this argument. The substitution $z = \tan \theta$ gives exactly

$$K(\sqrt{1 - s^2}) = \int_0^\infty \frac{dz}{\sqrt{(1 + z^2)(1 + s^2 z^2)}}.$$

The integral on $[0, 1]$ is at most 1, and that on $[1/s, \infty)$ is at most $\int_{1/s}^\infty dz/(sz^2) = 1$. On $[1, 1/s]$ the integrand is $z^{-1}(1 + z^{-2})^{-1/2}(1 + s^2 z^2)^{-1/2}$. For nonnegative a, b , $0 \leq 1 - (1 + a)^{-1/2}(1 + b)^{-1/2} \leq (a + b)/2$; this follows by applying $(1 + x)^{-1/2} \geq 1 - x/2$ separately and using $1 - AB = (1 - A) + A(1 - B)$ with $0 < A \leq 1$. The difference of the middle integral from $\int_1^{1/s} dz/z = L$ has magnitude at most

$$\frac{1}{2} \int_1^{1/s} (z^{-3} + s^2 z) dz = \frac{1 - s^2}{2} \leq \frac{1}{2}.$$

Consequently

$$L - \frac{1}{2} \leq K(\sqrt{1 - s^2}) \leq L + 2. \quad (\text{GRF24})$$

The defining E integral has $1 \leq E \leq \pi/2$ and $E \rightarrow 1$ as $s \rightarrow 0$, by dominated convergence to the integral of $\cos \theta$. By (GRF23),

$$\frac{sK}{E} = \frac{A}{B} \rightarrow 0. \quad (\text{GRF25})$$

Since $K \geq \pi/2$ and $E \leq \pi/2$, this implies $s \rightarrow 0$ and $L \rightarrow \infty$. Taking logarithms in (GRF25) gives

$$L = \log(B/A) + \log K - \log E = \log t + \log L + O(1).$$

Here (GRF24) gives $\log K = \log L + o(1)$ and $\log(B/A) = \log t + o(1)$. As $L \rightarrow \infty$, division of $\log t = L - \log L + O(1)$ by L proves $L \sim \log t$ and then $K \sim \log t$. Now (GRF23) yields

$$u \sim 2/\log t, \quad v \sim 2t, \quad s = O(1/(t \log t)). \quad (\text{GRF26})$$

The same proof with $A = 1$, $B = 1 + t$ is exactly the claimed statement for u_t at $(2/t, \pi/t)$.

The exact expression (GRF13) gives

$$\frac{\Sigma_\varepsilon}{2q} = A_\varepsilon \left[1 + \log \frac{1 + s_\varepsilon}{2u_\varepsilon} \right] + \frac{m_\varepsilon}{q} \log \frac{1 + s_\varepsilon}{1 - s_\varepsilon}. \quad (\text{GRF27})$$

For $0 < s \leq 1/2$, $0 \leq \log((1+s)/(1-s)) \leq 4s$, by integration of $2/(1-s^2)$. Thus the last term is $O(1/\log t) = o(1)$ by (GRF26). Also $\log(1+s) = o(1)$. Since $A_\varepsilon - 1 = O(1/q)$ and $1 + |\log u_\varepsilon| = O(1 + \log \log t)$, the error from replacing A_ε by 1 is $O((1 + \log \log t)/q) = o(1)$. The last conclusion follows from (GRF20): for large k , $\log t \leq q/R \leq q$, so $\log \log t = O(\log q) = o(q)$. Therefore

$$\frac{\Sigma_\varepsilon}{2q} = 1 + \log \frac{1}{2u_\varepsilon} + o(1), \quad \frac{1}{4} e^{-\Sigma_\varepsilon/(2q)} = \frac{u_\varepsilon}{2e} (1 + o(1)).$$

Since $1/e < 1$, this second quantity is eventually the smaller member in (GRF12), proving (GRF21).

We now verify every remaining finite guard and error. The support width satisfies $v_\varepsilon^2 - u_\varepsilon^2 \sim 4t^2$, so

$$m_\varepsilon(v_\varepsilon^2 - u_\varepsilon^2) \sim 2qt^3 \rightarrow \infty.$$

Furthermore

$$\frac{R}{qb_\varepsilon} \sim \frac{eR \log t}{q} \rightarrow 0,$$

which proves the cutoff guard. The exact parameters satisfy $\alpha_\varepsilon = O(1/t)$ and $\beta_\varepsilon = O(1/t)$, whence

$$\Lambda_\varepsilon = O((\log t)^2/t + \log t/t) = O((\log t)^2/t) = o(1).$$

Since $q/\log t \rightarrow \infty$, one has $qu_\varepsilon \rightarrow \infty$ and $r_{-, \varepsilon} \rightarrow c_\Gamma/\sqrt{2} > 0$. Taking logarithms in the exact (GRF17) gives

$$\begin{aligned} \log \eta_\varepsilon = & -q \log 8 + 4 \log m_\varepsilon + (2\varepsilon + 1/2) \log b_\varepsilon + \frac{3}{2} \Lambda_\varepsilon \\ & + \log \frac{32r_+}{r_{-, \varepsilon}(\varepsilon + 1/4)}. \end{aligned}$$

Here $\log m_\varepsilon = \log q + \log t + O(1) = o(q)$: $\log q = o(q)$ and (GRF20), with $R \rightarrow \infty$, gives $\log t = o(q)$. Also $\log b_\varepsilon = -\log \log t + O(1) = o(q)$, and all other displayed terms are $o(q)$. Both block errors are therefore $\exp[-q \log 8 + o(q)]$; their maximum has the same form and is eventually less than 1.

Finally $D_2 \leq qR^2$ by its root-sum representation and $M_4 \leq qR^4$. The two terms in (GRF18) are bounded by

$$\frac{R^2}{qb_\varepsilon^2} \left(1 + \frac{R^2/(q^2 b_\varepsilon^2)}{2(1 - R^2/(q^2 b_\varepsilon^2))} \right).$$

The parenthesis tends to 1 and $b_\varepsilon^{-2} = O((1 + \log t)^2)$. This proves the first estimate in (GRF22). Dividing it by q and using $R \log t/q \rightarrow 0$ proves $C_\chi = o(q)$. $\square$

1.6 Source pins and exact audit boundary

The received text was read in full before reconstruction. Its SHA256 is

b89f991d351b9a8b7cc2cf76768eb9997d79c50e8df03929fe82ea427e4a60ef.

The source file `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS.tex` in `output/Split_Zero_Web_Continuation_CURRENT` has SHA256

9926c5b5670777548f15f3fd626a4237d2c10a26834f51ba2b466a3b3a349656.

The identical immutable receiving copy is stored in `work/growing_ratio_math_integration_20260917/baseline_692`. The exact inputs inspected there are EIQ1–33, lines 11082–11539; OOQ7 and its complete proof, lines 13597–13641; ECL6, lines 18491–18505, and ECL14, lines 18664–18674; PCRP1–8, lines 57079–57169; RGC1–2, lines 57416–57447; and the earlier explicitly reference-only endpoint statement UCT40–42, lines 56636–56660. The related `02_FULL_GAMMA_PROOFS.tex` has SHA256

c226f7110ed7929b62c17b43bc6759d089a72e766fca387ebffcd80723a3c08;

its original mass and density declarations HMR19–20 are at lines 15889–15902. The complete EIQ provider is also present there at lines 1897–2344; its ECL provider begins at line 18675.

The direct elementary elliptic-integral proof in Theorem 4 does not rely on the “standard complete-elliptic expansions” cited by the earlier UCT40 paragraph. The evaluation proof independently reconstructs OOQ7 with the actual lower density and derivative constant, rather than using its earlier compact-parameter supremum. Thus no compact-parameter bound is extrapolated to $(\alpha_\varepsilon, \beta_\varepsilon) \rightarrow (0, 0)$. The two guards $m_\varepsilon \geq 1$ are essential to the displayed definitions; finite degrees $r < 3$ are outside this finite theorem, without affecting the eventually growing-ratio conclusion.

2 The reference norm at its actual moving parameters

The coordinate throughout is the literal coordinate $y = q\sqrt{x}$ on the positive half-line. The underlying measure is

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{GRE1})$$

It is not divided by its mass. Fix $q > 0$, an integer $m \geq 1$, and $\alpha, \beta > 0$. Write $\mu = \mu_{\alpha, \beta}$ for the probability measure constructed in the complete EIQ proof included in the accompanying complete source, with support $[a, b] = [u^2, v^2]$, $0 < u < v$, logarithmic potential $P(x) = \int \log|x - s| d\mu(s)$, and Robin constant λ in the convention

$$V(x) = \beta\sqrt{x} - \alpha \log x, \quad D(x) = V(x) - 2P(x) - \lambda \geq 0 \quad (x > 0). \quad (\text{GRE2})$$

Equality holds on $[a, b]$ and is strict outside it. The complete EIQ argument, included in the accompanying complete source, constructs the endpoints for every point of $(0, \infty)^2$, proves positivity and mass one, computes the Cauchy transform, and proves this inequality on both components of the exterior. In particular the inequality is available near zero as well as at infinity. Put

$$W = v^2 - u^2, \quad c = W/4, \quad w = (u + v)/2, \quad z = uv. \quad (\text{GRE3})$$

Here c is logarithmic capacity, and z is a positive scalar, not a complex integration variable.

2.1 Exact Robin and logarithmic-moment identities

The complete ECL1–14 proof is included in the accompanying complete source. Its moving-support derivative has the following precise test-function meaning:

$$\frac{\partial}{\partial \alpha} \int f(x) d\mu_{\alpha, \beta}(x) = \frac{1}{2} \int_a^b f(x) \left(1 - \frac{z}{x}\right) \frac{dx}{\pi \sqrt{(b-x)(x-a)}}. \quad (\text{GRE4})$$

It holds at fixed β for every continuously differentiable f on a neighbourhood of the support. To track the endpoint terms, write $p(x) = h(x)/(2\pi)$ and $\rho(x) = p(x)\sqrt{(b-x)(x-a)}$. Differentiation at the actual moving endpoints gives the signed density

$$\left[\dot{p}(x)\sqrt{(b-x)(x-a)} + \frac{p(x)\{\dot{b}(x-a) - \dot{a}(b-x)\}}{2\sqrt{(b-x)(x-a)}} \right] \mathbf{1}_{(a,b)}(x). \quad (\text{GRE5})$$

The proof changes variables to $x = a + (b-a)s$, $0 \leq s \leq 1$, differentiates there, and integrates the term containing f' by parts. The boundary term is proportional to $\sqrt{s(1-s)}$ and vanishes. The remaining derivative is integrable, bounded by a constant times $[s(1-s)]^{-1/2}$. Thus no endpoint atom is present and no moving-support term was dropped. The complete transform and zero-mass logarithmic-energy calculation in ECL7–10 identifies (GRE5) exactly with (GRE4).

In particular, setting $\mathcal{L} = P(0) = \int \log x d\mu(x)$ gives

$$\begin{aligned} \lambda &= 2(\alpha + 2) - 2\alpha \log w - 2 \log c, \\ \mathcal{L} &= 2(\alpha + 1) \log w - \alpha \log z - 2, \\ \lambda_\alpha &= -2 \log w, \quad \mathcal{L}_\alpha = 2 \log w - \log z. \end{aligned} \quad (\text{GRE6})$$

The constant -2 in the logarithmic moment remains. Direct addition, using $w^2/c = (u+v)/(v-u)$, yields

$$\boxed{2P(0) + \lambda = 2\alpha \left(1 + \log \frac{w}{z}\right) + 2 \log \frac{v+u}{v-u}}. \quad (\text{GRE7})$$

This is the received equation (6), with the original potential and Robin sign convention.

For clarity, the second Robin derivative also retains all support movement. For $\beta_0 > 0$ put $c_0 = \beta_0/\beta$. The dilation $x \mapsto c_0^2 x$ pushes μ_{α, β_0} to $\mu_{\alpha, \beta}$: on every probability measure its energy changes by the constant $-2(\alpha + 1) \log c_0$, so the proved unique minimizer is carried to the proved unique minimizer. At corresponding points its potential is $2 \log c_0 + P_{\alpha, \beta_0}(x)$, while its external field is $V_{\alpha, \beta_0}(x) - 2\alpha \log c_0$. Consequently

$$\lambda(\alpha, \beta) = \lambda(\alpha, \beta_0) + 2(\alpha + 2) \log(\beta/\beta_0), \quad \boxed{\lambda_\beta = \frac{2(\alpha + 2)}{\beta}}. \quad (\text{GRE8})$$

The distinct identity $\int \sqrt{x} d\mu = 2(\alpha + 1)/\beta$ concerns the moment, not the Robin derivative. It follows by differentiating the energy of the competitor $x \mapsto c_0^2 x$ at $c_0 = 1$ as in EIQ31.

2.2 Original Gamma constants, mass, and monic factorials

Retain exactly

$$c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \quad \mathfrak{r}(y) = \frac{\sigma(y)e^{\pi y/2}\sqrt{y}}{\sqrt{2}} \quad (y > 0). \quad (\text{GRE9})$$

The complete derivation of Binet's identity from the defining Gamma integral is reproduced in the accompanying complete source as BI1–26. Its remainder kernel is $K(s) = (\coth(s/2) - 2/s)/(2s)$ with $0 < K(s) < 1/12$. Therefore at $\zeta = 1/4 + iy/2$ its remainder satisfies $|R(\zeta)| \leq 1/3$, and twice the real part of that exact identity gives, with $Y = |y|$,

$$\sigma(y) = |\zeta|^{-1/2} \exp\{-Y \arctan(2Y) - 1/2 + 2\Re R(\zeta)\}. \quad (\text{GRE10})$$

For $Y > 0$, $Y \arctan(1/(2Y)) \leq 1/2$, whereas $\arctan(2Y) \leq \pi/2$; the same resulting inequalities hold at $Y = 0$. Thus the exponent in (GRE10) lies between $-\pi Y/2 - 7/6$ and $-\pi Y/2 + 2/3$. Also

$$\frac{(1+Y)^2}{4} - |\zeta|^2 = \frac{3+8Y}{16}, \quad |\zeta|^2 - \frac{(1+Y)^2}{20} = \frac{(4Y-1)^2}{80}.$$

Taking the appropriate positive fourth roots proves

$$c_\Gamma e^{-\pi|y|/2}(1+|y|)^{-1/2} \leq \sigma(y) \leq C_\Gamma e^{-\pi|y|/2}(1+|y|)^{-1/2}. \quad (\text{GRE11})$$

In particular all polynomial moments are finite and

$$r_+ = C_\Gamma/\sqrt{2}, \quad r_- = (c_\Gamma/\sqrt{2})\sqrt{\frac{qu}{1+qu}}, \quad \mathfrak{r}(y) \leq r_+, \quad \mathfrak{r}(q\sqrt{x}) \geq r_- \quad (a \leq x \leq b). \quad (\text{GRE12})$$

The lower estimate follows from the increasing function $y/(1+y)$.

For completeness, the mass and factorial monic norms can be recovered without selecting another measure. The beta identity, proved in BI13, with $a > 0$ and the substitution $x = \log(t/(1-t))$, gives

$$\frac{\Gamma(a+i\eta)\Gamma(a-i\eta)}{\Gamma(2a)} = \int_{\mathbb{R}} e^{i\eta x} (2 \cosh(x/2))^{-2a} dx. \quad (\text{GRE13})$$

The beta integrand after substitution is absolutely integrable. At $a = 1/4$, Fourier inversion therefore gives

$$\int_{\mathbb{R}} e^{i\xi y} d\sigma(y) = \sqrt{2\pi}(\cosh \xi)^{-1/2}. \quad (\text{GRE14})$$

Here Fourier inversion can be justified directly by inserting a Gaussian factor $e^{-\delta\eta^2}$ in the inversion integral, using Fubini, and evaluating the Gaussian Fourier integral. The result is convolution with the probability density $(4\pi\delta)^{-1/2}e^{-x^2/(4\delta)}$, which tends to the continuous beta density at each point: split at $|x| = \varepsilon$, use uniform continuity on the first piece and the Gaussian tail on the

second. On the Fourier side dominated convergence is valid by (GRE11), with $y = 2\eta$. This proves the inversion used in (GRE14). Both sides of (GRE14) extend holomorphically to $|\Im \xi| < \pi/2$: for the left side (GRE11) gives locally uniform exponential domination; for the right side the branch is positive at zero and has no zero of $\cosh \xi$ in the strip. The identity theorem consequently gives

$$\int e^{\theta y} d\sigma(y) = \sqrt{2\pi}(\cos \theta)^{-1/2} \quad (|\theta| < \pi/2). \quad (\text{GRE15})$$

This retains mass $\sqrt{2\pi}$ at $\theta = 0$.

Define monic polynomials p_n by the convergent generating series

$$F(y, z) = (1 + z^2)^{-1/4} e^{y \arctan z} = \sum_{n \geq 0} p_n(y) \frac{z^n}{n!}.$$

The coefficient of y^n in p_n is one, because $\arctan z = z + O(z^3)$ and the other prefactor has value one at zero. For sufficiently small real z, w , (GRE15) and $\cos(\arctan z + \arctan w) = (1 - zw)/\sqrt{(1 + z^2)(1 + w^2)}$ give the exact generating identity

$$\int F(y, z) F(y, w) d\sigma(y) = \sqrt{2\pi} (1 - zw)^{-1/2}.$$

Termwise differentiation at zero is justified by (GRE11) and exponential domination in a neighbourhood of $(z, w) = (0, 0)$. Hence

$$\int p_n p_j d\sigma = \mathbf{1}_{n=j} \sqrt{2\pi} n! (1/2)_n, \quad \gamma_n = \sqrt{2\pi} n! (1/2)_n = \sqrt{2\pi} \frac{(2n)!}{4^n}. \quad (\text{GRE16})$$

For a complex monic degree- n polynomial, subtraction of p_n leaves degree at most $n-1$, orthogonal to p_n . Pythagoras proves that γ_n is precisely the monic minimum; the product equality in (GRE16) follows by separating the even and odd factors of $(2n)!$.

2.3 A global density bound and a full tail integral

The actual positive density in EIQ10–11 obeys

$$0 < h(x) \leq \frac{\alpha}{xuv}, \quad \rho(x) = \frac{\sqrt{(v^2 - x)(x - u^2)}}{2\pi} h(x).$$

For $u^2 \leq x \leq v^2$, $(v^2 - x)(x - u^2) \leq v^2 x$. It follows, including the zero endpoint values, that

$$\|\rho\|_\infty \leq \mathcal{M} := \frac{\alpha}{2\pi u^2}. \quad (\text{GRE17})$$

This is the received equation (16).

Set, at these very parameters,

$$C_\lambda = 2 \log 2 + \max(\lambda, 0),$$

$$T_* = \max \left\{ 1, v, \left(\frac{8(\alpha + 2)}{\beta} \right)^2, \frac{4C_\lambda}{\beta} \right\}, \quad J_* = T_*^2 + \frac{8}{\beta^2}. \quad (\text{GRE18})$$

For $s \geq \max\{1, v\}$ and every support point y , $|s^2 - y| \leq s^2 + v^2 \leq 2s^2$. The logarithm is integrable even if $s = v$, so integration gives $P(s^2) \leq 2 \log s + \log 2$. Therefore

$$D(s^2) \geq \beta s - 2(\alpha + 2) \log s - C_\lambda.$$

For $s \geq 1$, $\log s \leq \sqrt{s}$: the maximum of $\log s/\sqrt{s}$ is $2/e < 1$, obtained by direct differentiation. If $s \geq T_*$, then $2(\alpha + 2) \log s \leq 2(\alpha + 2) \sqrt{s} \leq \beta s/4$ and $C_\lambda \leq \beta s/4$. Consequently

$$D(s^2) \geq \beta s/2 \quad (s \geq T_*), \quad \int_0^\infty e^{-mD(x)} dx \leq T_*^2 + \int_0^\infty 2s e^{-\beta s/2} ds = J_*. \quad (\text{GRE19})$$

The first interval uses $D \geq 0$, and the tail uses $m \geq 1$. The integral starts at zero. All constants are finite at each point of the full positive parameter quadrant.

2.4 Arcsine lower bound and explicit quantile upper bound

Define the actual monic minimum

$$h_{m,q}(\alpha, \beta) = \min_{p \in \mathbb{C}[x], p \text{ monic}, \deg p = m} \int_0^\infty |p(x)|^2 e^{-mV(x)} \mathfrak{r}(q\sqrt{x}) dx. \quad (\text{GRE20})$$

The integrals are finite: the weight near zero is at most $r_+ x^{m\alpha}$ and at infinity at most $r_+ x^{m\alpha} e^{-m\beta\sqrt{x}}$. Its density is positive on $(0, \infty)$, so its finite coefficient Gram matrix is positive definite. Completing the square in that finite matrix proves existence and uniqueness of the monic minimizer.

Let $d\nu = dx/(\pi\sqrt{(b-x)(x-a)})$ be arcsine probability on $[a, b]$. Its potential satisfies

$$\int \log|x - \zeta| d\nu(x) \geq \log c \quad (\zeta \in \mathbb{C}), \quad \int \log|x - y| d\nu(x) = \log c \quad (y \in [a, b]). \quad (\text{GRE21})$$

To prove both statements rootwise, put $C = (a+b)/2$, $d = W/2$ and $x = C + d\cos\theta$. The polynomial $Z^2 - 2(\zeta - C)Z/d + 1$ has roots z_1, z_2 with product one. On the unit circle, $|x - \zeta| = (d/2)|Z - z_1||Z - z_2|$. The circle mean of $\log|Z - z_j|$ equals $\log \max\{1, |z_j|\}$. For $|z_j| < 1$ this follows by integrating the absolutely convergent logarithmic series, for $|z_j| > 1$ by factoring $|z_j|$, and for $|z_j| = 1$ by a limit dominated near the single singular angle by a constant plus $|\log|\theta||$. The dependence on $\cos\theta$ identifies the whole-circle mean with the arcsine integral. Both roots have modulus one precisely when $\zeta \in [a, b]$, proving the equality; their two log-max terms are nonnegative, proving the inequality. All logarithmic integrals just used are absolutely integrable.

For a monic degree- m polynomial p , factorization and (GRE21) give $\int \log|p| d\nu \geq m \log c$. Integrate the Euler equality against ν . Fubini is justified by the bounded equilibrium density and the integrable arcsine logarithmic singularity, and gives

$$\int V d\nu = 2 \log c + \lambda. \quad (\text{GRE22})$$

The arcsine entropy is exactly

$$\int \log \frac{d\nu}{dx} d\nu = -\log(\pi c). \quad (\text{GRE23})$$

Indeed $\sqrt{(b-x)(x-a)} = (W/2)\sin\theta$, and $\pi^{-1} \int_0^\pi \log \sin\theta d\theta = -\log 2$. For the latter identity, put $I = \int_0^{\pi/2} \log \sin\theta d\theta$. Symmetry gives the same cosine integral, while doubling the argument in $\log \sin\theta + \log \cos\theta = \log \sin(2\theta) - \log 2$ gives $2I = I - (\pi/2)\log 2$. Endpoint logarithms are integrable.

Probability Jensen applied to $|p|^2 e^{-mV}/(d\nu/dx)$, whose logarithm is integrable, now proves

$$\int_a^b |p|^2 e^{-mV} dx \geq \exp \left\{ 2 \int \log|p| d\nu - m \int V d\nu - \int \log(d\nu/dx) d\nu \right\} \geq \pi c e^{-m\lambda}. \quad (\text{GRE24})$$

Multiplication by the lower remainder bound in (GRE12) gives the claimed lower monic bound.

For the upper bound take the unique midpoint quantiles $x_i \in (a, b)$ with $\mu((0, x_i]) = (i-1/2)/m$, $1 \leq i \leq m$, and define $p_m(x) = \prod_{i=1}^m (x - x_i)$. The distribution function of its empirical probability measure differs from that of μ by at most $1/(2m)$, and hence by at most $1/m$. For fixed $x > 0$ put

$$\delta = 1/m, \quad f_x(y) = \log \max\{|x - y|, \delta\}.$$

This is continuous and of bounded variation on $[a, b]$. If x lies inside $[a, b]$, its decreasing and increasing pieces each have variation at most $\log(1 + W/\delta)$. If x lies outside the interval it is monotone and its endpoint ratio after clipping is at most $1 + W/\delta$. Thus, in all cases, $\text{Var}_{[a,b]} f_x \leq 2 \log(1 + mW)$. Stieltjes integration by parts against the zero-mass empirical error has zero endpoint term, and gives

$$\sum_{i=1}^m f_x(x_i) \leq m \int f_x d\mu + 2 \log(1 + mW). \quad (\text{GRE25})$$

By (GRE17),

$$\int (f_x(y) - \log|x - y|) d\mu(y) \leq \mathcal{M} \int_{-\delta}^{\delta} \log(\delta/|s|) ds = 2\mathcal{M}\delta. \quad (\text{GRE26})$$

Each nodal logarithm is at most its clipped value; at a root its value is $-\infty$, where the resulting inequality is automatic. Combining (GRE25)–(GRE26), then multiplying by e^{-mV} , yields the global positive-half-line estimate

$$|p_m(x)|^2 e^{-mV(x)} \leq e^{4\mathcal{M}} (1 + mW)^4 e^{-m\lambda} e^{-mD(x)} \quad (x > 0). \quad (\text{GRE27})$$

Integrating it with (GRE12) and (GRE19) proves the complete finite norm theorem

$$\boxed{c_- e^{-m\lambda} \leq h_{m,q}(\alpha, \beta) \leq c_+ e^{-m\lambda}, \quad c_- = r_- \pi W / 4, \quad c_+ = r_+ J_* e^{4\mathcal{M}} (1 + mW)^4.} \quad (\text{GRE28})$$

This is the received (17). Neither its proof nor its constants require a compact parameter rectangle, an endpoint lower density estimate, or a lower bound on quantile-cell lengths.

2.5 Exact parity map, shifts, and factorial cancellation

Let $q \geq 1$ be an integer and $r = 2m + \varepsilon$ an integer with $m \geq 1$, $\varepsilon \in \{0, 1\}$. Define the full homogeneous reference monic minimum

$$\nu_r^0 = \min_{p \text{ monic, } \deg p = r} \int_{\mathbb{R}} y^{2q} |p(y)|^2 d\sigma(y). \quad (\text{GRE29})$$

An even positive weight makes the even and odd coefficient spaces orthogonal. For any monic degree- r polynomial, the component with parity opposite to r contributes a nonnegative separate squared norm and has no leading term, so the unique monic minimizer has parity r . The exact bijection between its parity fibre and the monic degree- m fibre is

$$p(y) = q^r (y/q)^\varepsilon P(y^2/q^2). \quad (\text{GRE30})$$

The leading coefficient is $q^{r-\varepsilon-2m} = 1$, and expansion in powers proves surjectivity and injectivity. Both half-lines contribute equally. Substitution $y = q\sqrt{x}$ has Jacobian $q/(2\sqrt{x})$, and (GRE9) gives exactly

$$\boxed{\nu_r^0 = \sqrt{2} q^{2(q+r)+1/2} h_{m,q}(\alpha, \beta), \quad \alpha = \frac{q + \varepsilon - 3/4}{m}, \quad \beta = \frac{\pi q}{2m}.} \quad (\text{GRE31})$$

Every mass and power, including the quarter shift, is retained.

Put $t = r/q$, $\alpha_0 = 2/t$, $\beta_0 = \pi/t$, and $\lambda_0 = \lambda(\alpha_0, \beta_0)$. The differences are exactly

$$A := m(\alpha - \alpha_0) = \varepsilon - \frac{3}{4} + \frac{q\varepsilon}{r}, \quad B := m(\beta - \beta_0) = \frac{\pi q\varepsilon}{2r}. \quad (\text{GRE32})$$

Let $(\alpha_\theta, \beta_\theta) = (\alpha_0, \beta_0) + \theta(\alpha - \alpha_0, \beta - \beta_0)$, $0 \leq \theta \leq 1$, and let w_θ be its actual midpoint $(u_\theta + v_\theta)/2$. All these parameter pairs are positive. The full support derivatives in (GRE6), (GRE8) and the fundamental theorem of calculus give

$$\begin{aligned} \Delta &:= m\lambda(\alpha, \beta) - \frac{r}{2}\lambda_0 \\ &= \int_0^1 \left[-2A \log w_\theta + B \frac{2(\alpha_\theta + 2)}{\beta_\theta} \right] d\theta - \frac{\varepsilon}{2}\lambda_0. \end{aligned} \quad (\text{GRE33})$$

Thus the finite, explicitly evaluable parameter bound is

$$|\Delta| \leq B_{\text{par}} := 2|A| \int_0^1 |\log w_\theta| d\theta + 2|B| \int_0^1 \frac{\alpha_\theta + 2}{\beta_\theta} d\theta + \frac{\varepsilon}{2}|\lambda_0|. \quad (\text{GRE34})$$

No fixed-endpoint differentiation is used in this formula.

Define the original profile in its exact Robin form

$$\psi(t) = (1+t)(1 - \log(1+t)) - \frac{t}{4}\lambda(2/t, \pi/t). \quad (\text{GRE35})$$

To identify it explicitly, let $\kappa \in (0, 1)$ be the EIQ modulus and $s = \sqrt{1 - \kappa^2}$. The endpoint equations at these actual parameters give

$$u_t K = 2, \quad v_t E = 2(1+t), \quad 1+t = \frac{E}{sK}, \quad w_t = \frac{(1+t)(1+s)}{E}, \quad c_t = \frac{(1+t)^2 \kappa^2}{E^2}.$$

Inserting these formulas into (GRE6),(GRE35), with no endpoint change, gives

$$\psi(t) = \log \frac{1+s}{E} + t \log \frac{\kappa}{E}. \quad (\text{GRE36})$$

This is precisely the original UCT26 profile.

Write $n = q + r$. BI26 evaluated at $2n$ and the recurrence $\Gamma(2n+1) = 2n\Gamma(2n)$ give

$$\log(2n)! = (2n + \tfrac{1}{2}) \log(2n) - 2n + \tfrac{1}{2} \log(2\pi) + R_{2n}, \quad 0 < R_{2n} < \frac{1}{24n}.$$

Combine this with (GRE16),(GRE31). Exact cancellation gives

$$\log \frac{\nu_r^0}{\gamma_{q+r}} = 2(q+r)(1 - \log(1+t)) + \log h_{m,q} - \tfrac{1}{2} \log(1+t) - \log(2\pi) - R_{2(q+r)}. \quad (\text{GRE37})$$

Equivalently, for $X = \log h_{m,q} + m\lambda$,

$$\boxed{\log \frac{\nu_r^0}{\gamma_{q+r}} = 2q\psi(t) + X - \Delta - \tfrac{1}{2} \log(1+t) - \log(2\pi) - R_{2(q+r)}, \quad \log c_- \leq X \leq \log c_+}. \quad (\text{GRE38})$$

The useful finite endpoints, before any asymptotic replacement, are also exactly

$$\log \frac{\sqrt{2}q^{2(q+r)+1/2}}{\gamma_{q+r}} - m\lambda + \log c_- \leq \log \frac{\nu_r^0}{\gamma_{q+r}} \leq \log \frac{\sqrt{2}q^{2(q+r)+1/2}}{\gamma_{q+r}} - m\lambda + \log c_+. \quad (\text{GRE39})$$

2.6 Uniform size of the retained errors

The elementary elliptic proof included in the accompanying complete source establishes, uniformly also along the segment in (GRE33),

$$u_\theta \sim \frac{2}{\log t}, \quad v_\theta \sim 2t, \quad W \sim 4t^2, \quad w_\theta \sim t, \quad |\lambda_0| + |\lambda| = O(1 + \log t) \quad (\text{GRE40})$$

as $q \rightarrow \infty$, $t \rightarrow \infty$. The segment has $\alpha_\theta = O(1/t)$ and $\beta_\theta \asymp 1/t$. Equations (GRE32)–(GRE34) consequently give $|\Delta| = O(1 + \log t)$, since $A = O(1)$ and $B = O(1/t)$. The same support estimates in the full finite constants give

$$\mathcal{M} = O((\log t)^2/t), \quad T_* = O(t^2), \quad J_* = O(t^4). \quad (\text{GRE41})$$

For T_* , its third entry is $O(t^2)$, its fourth is $O(t(1 + \log t)) = O(t^2)$, and its other entries are $O(t)$. These statements concern the actual endpoints, rather than compact rectangle extrema.

For the packet range of the received source, let $q = [1 + k(M-1)](k+1)^2$, with the fixed original multiplicity $M \geq 1$, $R_k = k\sqrt{\delta^2 + \gamma^2}$, and $\log t = o(q/R_k)$. Then $q/\log t \rightarrow \infty$, hence $qu \rightarrow \infty$ and r_- is bounded above and below by fixed positive constants. Thus $c_- \asymp t^2$, and

$$|\log c_-| + |\log c_+| = O(\log(q+r)). \quad (\text{GRE42})$$

Indeed $\mathcal{M} = O(1)$, $\log J_* = O(1 + \log t)$ and $mW = O(rt^2)$; since $q \geq 1$, $t = r/q \leq r$, so $\log(1 + rt^2) = O(\log(q + r))$. Lower bounds for J_* and $1 + mW$ are one, so this also bounds the absolute logarithm of c_+ , not only its positive part. The positive lower bound of r_+ is fixed. Similarly (GRE40) and $r_- \asymp 1$ bound both signs of $\log c_-$.

Finally (GRE38), (GRE34), (GRE42) and $0 < R_{2(q+r)} < 1/(24(q+r))$ prove

$$\boxed{\log \frac{\nu_r^0}{\gamma_{q+r}} = 2q\psi(r/q) + O(\log(q+r))}. \quad (\text{GRE43})$$

The constants are uniform along every such growing-ratio sequence and for either parity. This is the received equation (18). All finite content is already specified by (GRE18), (GRE28), (GRE33), and (GRE38)–(GRE39); no unspecified equilibrium norm theorem occurs.

3 Uniform elliptic endpoint estimates with the original quarter shift

Definitions and exact source identification. For $0 < s < 1$ define, by actual real integrals,

$$K_s = \int_0^{\pi/2} \frac{d\theta}{\sqrt{\cos^2 \theta + s^2 \sin^2 \theta}}, \quad E_s = \int_0^{\pi/2} \sqrt{\cos^2 \theta + s^2 \sin^2 \theta} d\theta. \quad (\text{GEL1})$$

Thus $K_s = K(\sqrt{1-s^2})$ and $E_s = E(\sqrt{1-s^2})$ in the modulus convention. A subscript s on K_s, E_s indicates the argument, not a derivative; derivatives are written d/ds . For $\alpha, \beta > 0$, let s solve

$$\frac{sK_s}{E_s} = \frac{\alpha}{\alpha+2}, \quad v = \frac{(\alpha+2)\pi}{\beta E_s}, \quad u = sv. \quad (\text{GEL2})$$

Retain the original coordinates and constants

$$a = u^2, \quad b = v^2, \quad W = b - a, \quad w = \frac{u+v}{2}, \quad c = \frac{W}{4}, \quad \lambda = 2(\alpha+2) - 2\alpha \log w - 2 \log c. \quad (\text{GEL3})$$

These are exactly EIQ2-5 and ECL3/6 of the frozen cumulative source 14_CURRENT_KERNEL_AND_MULTIPPLICITY_PROOFS.tex, whose checked SHA256 is

9926c5b5670777548f15f3fd626a4237d2c10a26834f51ba2b466a3b3a349656

In that source λ is denoted ℓ in EIQ/ECL. The statements below are proved for these functions from their displayed definitions; the source identifies them with the original equilibrium endpoints and Robin constant. No external asymptotic expansion is used.

Lemma 3.1 (Existence, derivative identities, and the logarithmic constant). *The equation for s in (GEL2) has exactly one solution in $(0, 1)$, smoothly depending on α . With $L = \log(4/s)$, as $s \downarrow 0$,*

$$K_s = L + O(s^2 L), \quad E_s = 1 + \frac{s^2}{2} \left(L - \frac{1}{2} \right) + O(s^4 L). \quad (\text{GEL4})$$

The constants in these bounds are absolute for $0 < s \leq 1/2$.

Proof. Differentiation under the integral sign, justified on each compact subinterval of $(0, 1)$, gives the first identity below. Integration of the derivative of $\sin \theta \cos \theta / (\cos^2 \theta + s^2 \sin^2 \theta)^{1/2}$ between its zero endpoint values gives the second:

$$\frac{dE_s}{ds} = \frac{s(K_s - E_s)}{1 - s^2}, \quad \frac{dK_s}{ds} = \frac{s^2 K_s - E_s}{s(1 - s^2)}. \quad (\text{GEL5})$$

For completeness the integration identity needed for the second is

$$\int_0^{\pi/2} \left\{ \frac{\cos^2 \theta - \sin^2 \theta}{D^{1/2}} + \frac{(1 - s^2) \sin^2 \theta \cos^2 \theta}{D^{3/2}} \right\} d\theta = 0, \quad D = \cos^2 \theta + s^2 \sin^2 \theta.$$

Indeed $dK_s/ds = -s \int \sin^2 \theta D^{-3/2} d\theta$; substitution in this zero integral gives the displayed formula. The integrals show $K_s > E_s > s^2 K_s$ for $0 < s < 1$. Therefore

$$\frac{d}{ds} \left(\frac{sK_s}{E_s} \right) = \frac{(K_s - E_s)(E_s - s^2 K_s)}{(1 - s^2)E_s^2} > 0.$$

As $s \downarrow 0$, the integrand of sK_s tends to zero for $\theta < \pi/2$ and is bounded by 1, so $sK_s \rightarrow 0$, while $E_s \rightarrow 1$. As $s \uparrow 1$, both integrals tend to $\pi/2$. Continuity, strict monotonicity, and the nonzero derivative prove existence, uniqueness, and smooth dependence.

Set $x = \cot \theta$ in the first integral of (GEL1); exactly

$$K_s = \int_0^\infty \frac{dx}{\sqrt{1+x^2}\sqrt{s^2+x^2}} = \operatorname{arsinh}(1/s) + A_s + B_s,$$

where

$$A_s = \int_0^1 \frac{(1+x^2)^{-1/2} - 1}{\sqrt{s^2+x^2}} dx, \quad B_s = \int_1^\infty \frac{dx}{\sqrt{1+x^2}\sqrt{s^2+x^2}}.$$

Both have convergent values at $s = 0$. The identity

$$\frac{d}{dx} \log \frac{x}{1 + \sqrt{1+x^2}} = \frac{1}{x\sqrt{1+x^2}}$$

gives

$$A_0 = \log \frac{2}{1 + \sqrt{2}}, \quad B_0 = \log(1 + \sqrt{2}), \quad A_0 + B_0 = \log 2.$$

The elementary inequalities

$$0 \leq 1 - (1+x^2)^{-1/2} \leq \frac{x^2}{2}, \quad 0 \leq \frac{1}{x} - \frac{1}{\sqrt{x^2+s^2}} \leq \min \left\{ \frac{1}{x}, \frac{s^2}{2x^3} \right\}$$

give, by splitting $A_s - A_0$ at $x = s$,

$$|A_s - A_0| \leq \frac{s^2}{4} + \frac{s^2}{4} \log(1/s), \quad |B_s - B_0| \leq \frac{s^2}{2} \int_1^\infty \frac{dx}{x^4} = \frac{s^2}{6}.$$

Furthermore $0 \leq \operatorname{arsinh}(1/s) - \log(2/s) = \log((1 + \sqrt{1+s^2})/2) \leq s^2/4$. This proves the first expansion, including its exact constant 4.

The bounds $1 \leq E_s \leq \pi/2$, the first expansion, and the first identity in (GEL5) show $0 \leq dE_s/ds \leq Cs \log(4/s)$ for $0 < s \leq 1/2$. Its integral is finite at zero, so integration from 0 is justified by first integrating from a positive lower endpoint and passing to zero. Consequently $E_s - 1 = O(s^2 L)$. Substituting this and the expansion of K_s back in the exact derivative identity improves it to

$$\frac{dE_s}{ds} = s(\log(4/s) - 1) + O(s^3 \log(4/s)).$$

The exact antiderivative calculation

$$\int_0^s z(\log(4/z) - 1) dz = \frac{s^2}{2} \left(\log(4/s) - \frac{1}{2} \right), \quad \int_0^s z^3 \log(4/z) dz = \frac{s^4}{4} \left(\log(4/s) + \frac{1}{4} \right)$$

proves the second expansion. $\square$

Theorem 3.2 (The whole moving parameter segment). *Let $q \rightarrow \infty$, $t = r/q \rightarrow \infty$, and $\varepsilon \in \{0, 1\}$, with $m = (r - \varepsilon)/2 > 0$. Define exactly*

$$\alpha = \frac{q + \varepsilon - 3/4}{m}, \quad \beta = \frac{\pi q}{2m}, \quad \alpha_0 = \frac{2}{t}, \quad \beta_0 = \frac{\pi}{t},$$

and for $0 \leq \theta \leq 1$ define $\alpha_\theta = (1 - \theta)\alpha_0 + \theta\alpha$ and $\beta_\theta = (1 - \theta)\beta_0 + \theta\beta$. Evaluate (GEL2)–(GEL3) at these parameters. All the following estimates hold uniformly in θ and ε :

$$L_\theta = \log t + \log \log t + \log 4 + o(1), \quad s_\theta \sim \frac{1}{t \log t}, \quad (\text{GEL6})$$

$$u_\theta \sim \frac{2}{\log t}, \quad v_\theta = 2t\{1 + O(1/t)\}, \quad (\text{GEL7})$$

$$W_\theta = 4t^2\{1 + O(1/t)\}, \quad w_\theta = t\{1 + O(1/t)\}, \quad (\text{GEL8})$$

$$\lambda(\alpha_\theta, \beta_\theta) = 4 - 4 \log t + O((\log t)/t). \quad (\text{GEL9})$$

In particular $\lambda = O(\log t)$ uniformly on the full segment.

Proof. Put $\delta = \varepsilon - 3/4$. The exact endpoint ratios are

$$\frac{\alpha}{\alpha_0} = \frac{1 + \delta/q}{1 - \varepsilon/r}, \quad \frac{\beta}{\beta_0} = \frac{1}{1 - \varepsilon/r}.$$

It follows uniformly on the segment that

$$\alpha_\theta = \frac{2}{t}\{1 + O(1/q)\}, \quad \beta_\theta = \frac{\pi}{t}\{1 + O(1/r)\}, \quad \frac{\alpha_\theta}{\alpha_\theta + 2} = \frac{1}{t+1}\{1 + O(1/q)\}.$$

The right side tends uniformly to zero. The strict monotonicity and positive values of sK_s/E_s proved above imply $s_\theta \rightarrow 0$ uniformly. Hence (GEL4) and (GEL2) give

$$s_\theta L_\theta = \frac{1}{t}\{1 + O(1/t + 1/q + s_\theta^2 L_\theta)\} = \frac{1}{t}\{1 + o(1)\}.$$

Since $s_\theta = 4e^{-L_\theta}$, taking logarithms gives

$$L_\theta - \log L_\theta = \log t + \log 4 + o(1).$$

Here $L_\theta \rightarrow \infty$ uniformly. Dividing by L_θ first shows $L_\theta/\log t \rightarrow 1$ uniformly. Substitution then gives $\log L_\theta = \log \log t + o(1)$ and proves (GEL6).

The original endpoint equation, without changing coordinates, gives

$$u_\theta = \frac{\pi \alpha_\theta}{\beta_\theta K_{s_\theta}} = \frac{2\{1 + O(1/q)\}}{L_\theta\{1 + O(s_\theta^2)\}} \sim \frac{2}{\log t}.$$

The other endpoint satisfies

$$\frac{v_\theta}{2t} = \frac{1 + \alpha_\theta/2}{1 + O(1/r)}\{1 + O(s_\theta^2 L_\theta)\} = 1 + O(1/t),$$

because $r = tq$ and $s_\theta^2 L_\theta \sim 1/(t^2 \log t)$. Now $u_\theta/v_\theta = s_\theta$, so $W_\theta = v_\theta^2(1 - s_\theta^2)$ and $w_\theta = v_\theta(1 + s_\theta)/2$ prove (GEL8). Thus $\log w_\theta = \log t + O(1/t)$ and $\log(W_\theta/4) = 2\log t + O(1/t)$. Substitution in the exact definition of λ , with $\alpha_\theta = O(1/t)$, proves (GEL9). $\square$

Proposition 3.3 (Exact Robin derivatives and the full quarter-shift cost). *For every $\alpha, \beta > 0$,*

$$\lambda_\alpha = -2 \log w, \quad \lambda_\beta = \frac{2(\alpha + 2)}{\beta}. \quad (\text{GEL10})$$

On the segment of the theorem, these imply uniformly

$$\lambda_\alpha = -2 \log t + O(1/t), \quad \lambda_\beta = \frac{4t}{\pi} + O(1).$$

In the original shifted parameters, including the change of degree,

$$m\lambda(\alpha, \beta) - \frac{r}{2}\lambda(2/t, \pi/t) = \frac{3}{2}\log t + O((\log t)/t). \quad (\text{GEL11})$$

The exact second derivatives are

$$\lambda_{\alpha\alpha} = -\frac{(1-s)^2 K_s(E_s - sK_s)}{(K_s - E_s)(E_s - s^2 K_s)}, \quad \lambda_{\alpha\beta} = \frac{2}{\beta}, \quad \lambda_{\beta\beta} = -\frac{2(\alpha + 2)}{\beta^2}. \quad (\text{GEL12})$$

They are respectively $-1 + O(1/\log t)$, $O(t)$, and $O(t^2)$ uniformly on that segment.

Proof. Set $Q = E_s - sK_s > 0$. Solving the endpoint relation exactly gives

$$\alpha = \frac{2sK_s}{Q}, \quad v = \frac{2\pi}{\beta Q}, \quad w = \frac{\pi(1+s)}{\beta Q}, \quad c = \frac{\pi^2(1-s^2)}{\beta^2 Q^2}.$$

The derivative identities already proved give

$$Q' = \frac{E_s - K_s}{1+s}, \quad \alpha' = \frac{2(K_s - E_s)(E_s - s^2 K_s)}{(1-s^2)Q^2} > 0, \quad \frac{d \log w}{ds} = \frac{(1-s)K_s}{(1+s)Q}$$

at fixed β . Direct differentiation of c gives

$$\frac{d \log c}{ds} = \frac{2\{(1-s+s^2)K_s - E_s\}}{(1-s^2)Q}.$$

These expressions imply $\alpha(d \log w/ds) + d \log c/ds = \alpha'$. To check every term, multiplication by $(1-s^2)Q^2/2$ reduces this identity to

$$s(1-s)^2 K_s^2 + \{(1-s+s^2)K_s - E_s\}(E_s - sK_s) = (K_s - E_s)(E_s - s^2 K_s),$$

whose two sides expand to $-s^2 K_s^2 + (1+s^2)E_s K_s - E_s^2$. Differentiating the definition of λ with respect to s now gives $d\lambda/ds = -2 \log w \alpha'$, proving the first identity in (GEL10). At fixed α , s is fixed, $w \propto \beta^{-1}$, and $c \propto \beta^{-2}$, which proves the second. The derivative estimates follow from the theorem and $\alpha_\theta = 2/t\{1 + O(1/q)\}$, $\beta_\theta = \pi/t\{1 + O(1/r)\}$; in particular the beta error is $O(1+t/r) = O(1)$.

Write $\lambda_0 = \lambda(2/t, \pi/t)$. The exact parameter differences are

$$m(\alpha - 2/t) = \delta + \varepsilon/t, \quad m(\beta - \pi/t) = \frac{\pi\varepsilon}{2t}, \quad m - \frac{r}{2} = -\frac{\varepsilon}{2}.$$

The fundamental theorem of calculus on the actual straight segment gives

$$\begin{aligned} m(\lambda - \lambda_0) &= \left(\delta + \frac{\varepsilon}{t}\right) \int_0^1 \lambda_\alpha(\alpha_\theta, \beta_\theta) d\theta + \frac{\pi\varepsilon}{2t} \int_0^1 \lambda_\beta(\alpha_\theta, \beta_\theta) d\theta \\ &= -2\delta \log t + 2\varepsilon + O((\log t)/t). \end{aligned}$$

Subtracting $\varepsilon\lambda_0/2$ and using $\lambda_0 = 4 - 4 \log t + O((\log t)/t)$ yields (GEL11), since $-2\delta + 2\varepsilon = 3/2$. Finally differentiating λ_α using the displayed values of $d \log w/ds$ and α' gives (GEL12); the other two derivatives follow directly from (GEL10). The expansions give $Q = 1 + O(sL)$ and $K_s/(K_s - E_s) = 1 + O(1/L)$, proving the stated uniform Hessian bounds. $\square$

Theorem 3.4 (The large-ratio profile from the real integrals). *For each $t > 0$, let s be determined by $1+t = E_s/(sK_s)$ and put $\kappa = \sqrt{1-s^2}$. Define the exact UCT26 profile*

$$\psi(t) = \log \frac{1+s}{E_s} + t \log \frac{\kappa}{E_s}.$$

Then, with $L = \log(4/s)$,

$$\psi(t) = \frac{s}{2} - \frac{s}{4L} + O(s^3 L), \quad \psi(t) \sim \frac{1}{2t \log t} \quad (t \rightarrow \infty). \quad (\text{GEL13})$$

Proof. The exact logarithmic identity $\log((1+s)/\kappa) = \frac{1}{2} \log((1+s)/(1-s))$ gives

$$\psi(t) = \operatorname{arctanh} s + \frac{E_s}{sK_s} \log \frac{\kappa}{E_s}.$$

Taylor's theorem for $\log(1+x)$, with its remainder bounded on a fixed neighbourhood of zero, and (GEL4) give

$$\log E_s = \frac{s^2}{2} \left(L - \frac{1}{2} \right) + O(s^4 L^2), \quad \log \kappa = -\frac{s^2}{2} + O(s^4),$$

so

$$\log \frac{\kappa}{E_s} = -\frac{s^2}{2} \left(L + \frac{1}{2} \right) + O(s^4 L^2), \quad \frac{E_s}{sK_s} = \frac{1}{sL} \{1 + O(s^2 L)\}.$$

Their product is $-s/2 - s/(4L) + O(s^3 L)$. Since $\operatorname{arctanh} s = s + O(s^3)$, this proves the first formula. The reference parameters are exactly $\alpha_0 = 2/t, \beta_0 = \pi/t$, so (GEL6) gives $s \sim 1/(t \log t)$ and $L \rightarrow \infty$; the remainder divided by s tends to zero because $s^2 L \rightarrow 0$. This proves the stated leading equivalent. $\square$

The original root coordinate. The foregoing results need only $q \rightarrow \infty$ and $t \rightarrow \infty$. If in addition q is comparable to k^2 or k^3 and $\log t = o(q/k)$, then in the original coordinate $y = q\sqrt{x}$, uniformly on the whole parameter segment,

$$\frac{qu_\theta}{k} \sim \frac{2q}{k \log t} \longrightarrow \infty.$$

Consequently for every fixed constant $C > 0$ and every root-radius sequence $0 \leq R_k \leq Ck$, one has $R_k/(qu_\theta) \rightarrow 0$ uniformly. This is an explicit consequence in the original coordinate, retaining both the factor q and the actual endpoint u_θ .

4 Original arithmetic transfer and the complete shifted action

This section proves the arithmetic receiving calculation for the moving equilibrium comparison. It retains the stipulated complete zero quartet, all source masses and divided jets, the original remainder quotient, and both adjacent contraction factors. The moving-cutoff full-form theorem and the moving-parameter reference-norm theorem are the analytic results proved in the companion sections. Their exact conclusions are displayed where used; none of their constants is replaced by a compact-parameter constant. The new calculation here is the exact arithmetic transfer, its degree-uniform asymptotic consequences, and its original complete-action receiver.

4.1 Objects, coefficient maps, and all-degree arithmetic constants

Fix the original complete quartet data

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m \in \mathbb{N}_{>0}, \quad h(z) = \prod_{\zeta \in \{1/2 \pm \delta \pm i\gamma\}} (z - \zeta)^m, \quad v_h = 2\xi/h.$$

The zero order at each centre is exactly the stated m , so the entire quotient v_h is nonzero at every centre. Along the original tensor subsequence put

$$\begin{aligned} k = 4\ell + 1 \geq 5, \quad c = k/2, \quad e_k = 1 + k(m-1), \quad q = e_k(k+1)^2, \quad R_k = k\sqrt{\delta^2 + \gamma^2}, \\ \lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^{e_k}, \\ Q(y) = i^{-q}\chi(c + iy), \quad E = \mathbb{C}[S]/(\chi), \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ d\mu_k(y) = w_h^{*k}(y) dy, \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \end{aligned} \tag{GRA1}$$

The masses are $(\int w_h)^k$ and $\sqrt{2\pi}$, respectively. For $P \in \mathcal{P}_j(S) = \mathbb{C}[S]_{\leq j}$ the arithmetic squared norm is $\|P\|_{\text{ar}}^2 = \int |P(c + iy)|^2 d\mu_k(y)$; the corresponding source Gamma norm uses $d\sigma$ in the same coordinate. Inner products are conjugate-linear in their first argument. Every polynomial coefficient below may be complex.

The precise coefficient maps are

$$U_j P(y) = i^{-j} P(c + iy), \quad U_j^{-1} p(S) = i^j p((S - c)/i), \quad U_{q+j} M_\chi = M_Q U_j. \tag{GRA2}$$

Direct substitution proves both inverse identities and the multiplication identity. The leading coefficient of a polynomial of degree j is unchanged. The leading coefficient of a polynomial of degree $a < j$ is multiplied by i^{a-j} , which is retained. Thus U_j bijects the two degree- j monic affine spaces, and its scalar phase has modulus one in each pulled-back integral norm. No Euclidean norm of a coefficient vector is asserted to be invariant under translation.

Write $\alpha_* = \pi/2$ to distinguish this fixed tail constant from the moving equilibrium parameters. The original AMT/TW constants are

$$\begin{aligned} p_h = 8m - \frac{9}{2}, \quad B_h = 42 + 8m, \quad M_h = 21 + 4m, \\ \vartheta_h = \int_{-1}^1 w_h(y) dy, \quad M_{\alpha_*} = \int_{\mathbb{R}} e^{\alpha_* y} w_h(y) dy, \\ c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha_*(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p_h+1} M_{\alpha_*}^{k-1}, \\ D_d = [1 + 4(d + M_h)^2]^{M_h}, \quad L_{k,d} = \log(A_k D_d / a_k). \end{aligned} \tag{GRA3}$$

Here c_h and C_h^{tilt} are exactly the positive original TW7 convolution lower-envelope and BT12 compact/tail constants. Their already proved defining estimates, with the full numerator $2\xi/h$, are

$$\begin{aligned} w_h^{*k}(y) &\geq c_h \vartheta_h^{k-3} e^{-\alpha_*(|y|+k-3)} (k-2+|y|)^{-B_h}, \\ w_h^{*k}(y) &\leq k C_h^{\text{tilt}} M_{\alpha_*}^{k-1} e^{-\alpha_*|y|} (1+|y|/k)^{-p_h}, \\ c_\Gamma e^{-\alpha_*|y|} (1+|y|)^{-1/2} &\leq d\sigma/dy \leq C_\Gamma e^{-\alpha_*|y|} (1+|y|)^{-1/2}. \end{aligned} \quad (\text{GRA4})$$

These bounds hold for every convolution order $k \geq 3$. In particular there is no upper polynomial-degree restriction in the following argument.

The first and third estimates, $k-2+|y| \leq (k-2)(1+|y|)$, and $(1+|y|)^{B_h} \leq 2^{B_h/2}(1+y^2)^{M_h}$ give $w_h^{*k} \geq a_k(1+y^2)^{-M_h} d\sigma/dy$. The second and third, together with $1+|y|/k \geq (1+|y|)/k$ and $p_h > 1/2$, give $w_h^{*k} \leq A_k d\sigma/dy$. For the original Gamma orthonormal polynomials e_n one has

$$ye_n = \sqrt{(n+1)(n+1/2)}e_{n+1} + \sqrt{n(n-1/2)}e_{n-1}.$$

The two shifts on polynomials of degree at most d have norms at most $d+1$ and d , respectively. Hence $\|yf\|_\sigma \leq 2(d+1)\|f\|_\sigma$. Applying this inequality successively gives, for $0 \leq j \leq M_h$, $\|y^j f\|_\sigma^2 \leq [4(d+M_h)^2]^j \|f\|_\sigma^2$. Expansion of $(1+y^2)^{M_h}$ therefore bounds its weighted squared norm by $D_d \|f\|_\sigma^2$. Cauchy–Schwarz gives

$$\|f\|_\sigma^4 \leq \left(\int |f|^2 (1+y^2)^{-M_h} d\sigma \right) \left(\int |f|^2 (1+y^2)^{M_h} d\sigma \right).$$

Combining these inequalities and using the bijective substitution $S = c+iy$ proves, on the complete coefficient space and at every $d \geq 0$,

$$\frac{a_k}{D_d} H_d^\sigma \preceq H_d^{\text{ar}} \preceq A_k H_d^\sigma. \quad (\text{GRA5})$$

Both forms are positive definite by their positive densities, and all moments are finite by their exponential upper bounds. Applying (GRA5) to a nonzero vector proves $L_{k,d} \geq 0$. Its dependence on degree is explicit. Subtracting the logarithms in (GRA3) gives

$$\begin{aligned} \log(A_k/a_k) &= k\mathfrak{s}_h + (p_h+1)\log k + B_h \log(k-2) + \mathfrak{c}_h, \\ \mathfrak{s}_h &= \alpha_* + \log(M_{\alpha_*}/\vartheta_h), \\ \mathfrak{c}_h &= -\log M_{\alpha_*} + 3\log \vartheta_h - 3\alpha_* + \log \frac{C_h^{\text{tilt}} C_\Gamma 2^{B_h/2}}{c_h c_\Gamma}, \\ L_{k,d} &= O_h(k + \log(d+1)). \end{aligned} \quad (\text{GRA6})$$

The constant implicit here is independent of d and k .

4.2 Full forms, fixed affine fibres, and arithmetic monic minima

For $p \in \mathcal{P}_r(y)$ retain the two full forms

$$\mathcal{N}_0[p] = \int y^{2q} |p(y)|^2 d\sigma(y), \quad \mathcal{N}_\chi[p] = \int |Q(y)p(y)|^2 d\sigma(y).$$

The companion moving-cutoff theorem gives, with its actual finite constants $C_\chi(k, r)$ and $\eta(k, r)$ and all its displayed guards,

$$e^{-C_\chi}(1-\eta)\mathcal{N}_0[p] \leq \mathcal{N}_\chi[p] \leq (1+\eta)\mathcal{N}_0[p], \quad 0 \leq \eta < 1. \quad (\text{GRA7})$$

It is the original Q on the entire real line. In particular this theorem does not remove the inner integration interval. Applying (GRA5) at $d = q+r$ to $\chi U_r^{-1}p$ and using (GRA2) proves the exactly typed form of the arithmetic receiver:

$$\frac{a_k}{D_{q+r}} e^{-C_\chi}(1-\eta)\mathcal{N}_0[p] \leq \|\chi U_r^{-1}p\|_{\text{ar}}^2 \leq A_k(1+\eta)\mathcal{N}_0[p]. \quad (\text{GRA8})$$

Indeed the middle integrand after substitution is exactly $|Q(y)p(y)|^2$. This specifies the coordinate implicit in the received notation $\|\chi p\|_{\text{ar}}$.

For completeness the same constants act on every unchanged affine coefficient fibre. Let $C : \mathcal{P}_r(y) \rightarrow F$ be a specified complex linear map and $z \in \text{im } C$. Each positive form has a unique minimum on $Cp = z$: after choosing one lift, minimization is a strictly positive quadratic problem on $\ker C$. Apply each inequality of (GRA8) to every vector of this same nonempty fibre and take the infima. This proves the two inequalities for the resulting minimum forms, also at $z = 0$. In original coordinates the observation is CU_r , and U_r is the exact fibre bijection. Restriction follows by applying the form inequality to every vector in the specified subspace. No mixed matrix entry is set equal to zero by either operation.

Let ω_d^{ar} be the degree- d monic arithmetic squared norm, and retain the original Gamma monic squared norm

$$\gamma_d = \sqrt{2\pi} d!(1/2)_d = \sqrt{2\pi} (2d)!/4^d.$$

The mass is its degree-zero value; the Gamma recurrence gives $\gamma_d/\gamma_{d-1} = d(d-1/2)$. The monic bijection U_d and (GRA5) give

$$\frac{a_k}{D_d} \gamma_d \leq \omega_d^{\text{ar}} \leq A_k \gamma_d. \quad (\text{GRA9})$$

Define the original relation minima and ratios by

$$\nu_r^{\text{ar}} = \min_{P(S) \text{ monic, deg } P=r} \|\chi P\|_{\text{ar}}^2, \quad \nu_r^0 = \min_{p(y) \text{ monic, deg } p=r} \mathcal{N}_0[p], \quad T_r = \nu_r^{\text{ar}}/\omega_{q+r}^{\text{ar}}.$$

Taking these identical monic affine minima in (GRA8), then dividing by the opposite endpoints of (GRA9), proves

$$\begin{aligned} \log T_r &\geq \log(\nu_r^0/\gamma_{q+r}) - L_{k,q+r} - C_\chi + \log(1 - \eta), \\ \log T_r &\leq \log(\nu_r^0/\gamma_{q+r}) + L_{k,q+r} + \log(1 + \eta). \end{aligned} \quad (\text{GRA10})$$

The denominator is the actual arithmetic monic minimum throughout.

Here are the complete finite scalar endpoints obtained from the companion reference theorem. Write $r = 2n_r + \varepsilon_r$ with $\varepsilon_r \in \{0, 1\}$, and retain its actual moving parameters, support, Robin constant λ_r , and positive constants $c_{-,r}$, $c_{+,r}$. That theorem proves

$$c_{-,r} e^{-n_r \lambda_r} \leq h_{n_r, q} \leq c_{+,r} e^{-n_r \lambda_r}, \quad \nu_r^0 = \sqrt{2} q^{2(q+r)+1/2} h_{n_r, q}.$$

Consequently, with every constant evaluated at this r ,

$$\begin{aligned} Z_{q,r} &= \log \frac{\sqrt{2} q^{2(q+r)+1/2}}{\gamma_{q+r}} - n_r \lambda_r, \\ \underline{E}_r &= Z_{q,r} + \log c_{-,r} - L_{k,q+r} - C_\chi(k, r) + \log(1 - \eta(k, r)), \\ \overline{E}_r &= Z_{q,r} + \log c_{+,r} + L_{k,q+r} + \log(1 + \eta(k, r)), \\ \underline{E}_r &\leq \log T_r \leq \overline{E}_r. \end{aligned} \quad (\text{GRA11})$$

These are usable finite endpoints wherever the analytic theorem's finite guards hold. They are obtained by inequalities, not by replacing an arithmetic norm with an equilibrium norm.

4.3 Original quotient, full-jet support, and adjacent allowance

For $N \geq q - 1$ the literal remainder map $J_N : \mathcal{P}_N(S) \rightarrow E$ is onto and has kernel $\chi \mathcal{P}_{N-q}(S)$; at $N = q - 1$ that kernel is zero. In the coefficient frame $1, S, \dots, S^N$ define its quotient minimum Gram and determinant by

$$G_N = (J_N(H_N^{\text{ar}})^{-1} J_N^*)^{-1}, \quad V_N = \det G_N.$$

To verify the precise ratio, use the frame

$$(\chi, \chi S, \dots, \chi S^{N-q}; 1, S, \dots, S^{q-1}).$$

Reordering it by degree makes a unit-triangular coefficient matrix, because χ is monic; its determinant therefore has modulus one. Positive block elimination gives the first equality below. Gram–Schmidt on the full source and on the relation coefficient space with norm $\|\chi P\|_{\text{ar}}$ gives the next two:

$$\det H_N^{\text{ar}} = \det H_{\text{rel},N} \det G_N, \quad \det H_N^{\text{ar}} = \prod_{d=0}^N \omega_d^{\text{ar}}, \quad \det H_{\text{rel},N} = \prod_{j=0}^{N-q} \nu_j^{\text{ar}}.$$

Empty products are one. Dividing consecutive equalities proves

$$d_{q+r} := \frac{V_{q+r}}{V_{q+r-1}} = T_r^{-1}, \quad r \geq 0. \quad (\text{GRA12})$$

This is a source/relation determinant identity for the original quotient $\mathbb{C}[S]/(\chi)$.

The actual support map remains as follows. Set $E_h = \mathbb{C}[z]/(h)$, $v_h = j_h v_h \in E_h^\times$, and $A_\Sigma = \sum_{i=1}^k M_{z_i}$ on $E_h^{\otimes k}$. Then

$$\eta_{h,k} : E \longrightarrow E_h^{\otimes k}, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_\Sigma) 1, \quad \eta_{h,k} M_S = A_\Sigma \eta_{h,k}. \quad (\text{GRA13})$$

In each local factor the nilpotent has exact order m . In each local tensor block their sum has exact order $1 + k(m-1)$: its power $k(m-1)$ has nonzero coefficient $[k(m-1)!]/[(m-1)!]^k$ on the product of the top nilpotent powers, and its next power vanishes because at least one exponent is m . The real and imaginary signs at the four original centres can be chosen independently, so every pair (a, b) occurs as a sum centre λ_{ab} . Distinct pairs give distinct centres because $\delta, \gamma > 0$. The annihilator of 1 under A_Σ is therefore exactly χ . Multiplication by the full unit preserves this annihilator, proving injectivity of (GRA13); direct multiplication proves its intertwining identity. The original source realization and jets consequently retain their exact commuting formula

$$\mathcal{V}(P) = P\left(\sum_i D_i\right) F_h^{\otimes k}, \quad J^{(k)} \mathcal{V}(P) = \eta_{h,k} J_N P.$$

In particular the relation norm above is measured before its quotient image is zero. No quotient by y^q enters this map.

Here is the complete adjacent calculation in these same norms. Let $p_n(S) = i^n Q_n((S-c)/i)$ be the original monic arithmetic orthogonal polynomial, let $b_n = [p_n]_\chi$, and put

$$R_N = G_N^{-1} = \sum_{n=0}^N b_n b_n^* / \omega_n^{\text{ar}}, \quad \beta_n = \omega_n^{\text{ar}} / \omega_{n-1}^{\text{ar}}.$$

Evenness of w_h , hence of its convolution, gives $Q_n(-y) = (-1)^n Q_n(y)$. Multiplication by S has the original recurrence $S p_n = p_{n+1} + c p_n - \beta_n p_{n-1}$, with the last term absent at zero. Inserting it into R_N cancels all interior adjacent pairs and gives

$$M_S R_N + R_N M_S^* - k R_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N^{\text{ar}}}.$$

Reflection $P(S) \mapsto P(k-S)$ permutes the roots of the even-degree polynomial χ and induces an involution Σ on E . Reflection is an isometry on every original remainder fibre, so the unique minimum implies $\Sigma^* G_N \Sigma = G_N$. Since $\Sigma b_n = (-1)^n b_n$, it follows that $b_N^* G_N b_{N+1} = 0$. For $x = G_N^{1/2} b_N$ and $y = G_N^{1/2} b_{N+1}$ the relative Hermitian control has nonzero eigenvalues $\epsilon_N, -\epsilon_N$, with

$$\epsilon_N^2 = \|x\|^2 \|y\|^2 / (\omega_N^{\text{ar}})^2.$$

The determinant lemma for adding and removing the last covariance column proves

$$\frac{b_N^* G_N b_N}{\omega_N^{\text{ar}}} = 1 - d_N \quad (N \geq q), \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}^{\text{ar}}} = d_{N+1}^{-1} - 1.$$

At $N = q - 1$ the first q monic remainder columns form a basis and the first fraction is exactly one. Thus no inverse at $q - 2$ is used. Together with (GRA12) these identities prove

$$\epsilon_{q+r-1}^2 = \beta_{q+r}(T_r - 1) \begin{cases} 1, & r = 0, \\ 1 - T_{r-1}^{-1}, & r \geq 1. \end{cases} \quad (\text{GRA14})$$

Only the total pairing vanished. In the original orthogonal kernel–boundary split, $x = x_K \oplus x_B$, $y = y_K \oplus y_B$, one has $x_B^* y_B = -x_K^* y_K$. The mixed exterior term therefore remains

$$\|x_K \otimes y_B - y_K \otimes x_B\|^2 = \|x_K\|^2 \|y_B\|^2 + \|x_B\|^2 \|y_K\|^2 + 2|x_K^* y_K|^2. \quad (\text{GRA15})$$

The unchanged exterior benchmark is also explicit. Let E_+ be the sum of the complete primary blocks with $a > k/2$. Since k is odd, $\dim E_+ = q/2$. The nonzero top wedge of a full primary basis has additive M_S eigenvalue $\text{Tr}(M_S|_{E_+})$; all nilpotent parts have trace zero. The relative Hermitian control's additive operator on exterior degree $q/2$ has largest eigenvalue ϵ_N , since it sums distinct members of $\{\epsilon_N, -\epsilon_N, 0, \dots, 0\}$. Its expectation on the stated wedge is $2 \text{Re} \text{Tr}(M_S|_{E_+}) - kq/2$. The wedge stays nonzero under the injective full-jet map (GRA13). Summing the positive odd integers $2a - k$ gives $(k + 1)^2/4$. Therefore

$$0 < \mathcal{L}_{h,k} := 2\delta e_k(k + 1) \frac{(k + 1)^2}{4} = \frac{\delta q(k + 1)}{2} \leq \epsilon_N \quad (N \geq q - 1). \quad (\text{GRA16})$$

Positive covariance gives $0 < d_n \leq 1$. Equation (GRA14) and the strict positive benchmark first force $d_q < 1$ and then $d_n < 1$ at every subsequent degree. Thus all logarithms below are defined for the actual objects.

Finally (GRA9) gives the sharpened finite recurrence enclosure

$$-L_{k,d} \leq \log \frac{\beta_d}{d(d - 1/2)} \leq L_{k,d-1}, \quad d \geq 1. \quad (\text{GRA17})$$

For its lower bound divide the lower bound for ω_d^{ar} by the upper bound for ω_{d-1}^{ar} ; reverse the choices for the upper bound. This is an adjacent norm comparison, not a derivative of a determinant asymptotic.

4.4 Growing ratios and uniformity on every shifted index

The companion moving-parameter reference theorem proves

$$\log(\nu_r^0 / \gamma_{q+r}) = 2q\psi(r/q) + O_h(\log(q + r)), \quad \psi(t) \sim \frac{1}{2t \log t}.$$

The companion full-form theorem proves, throughout $t = r/q \rightarrow \infty$, $\log t = o(q/R_k)$, that its finite guards eventually hold and that $C_\chi = O_h(R_k^2(1 + \log t)^2/q)$ and $\eta = \exp[-q \log 8 + o(q)]$. The constants in the reference estimate are those evaluated at the actual moving support. Insert these proved estimates into (GRA10); for $0 \leq \eta \leq 1/2$, $|\log(1 \pm \eta)| \leq 2\eta$. This proves

$$\log T_r = 2q\psi(t) + O_h \left(k + \log(q + r) + \frac{R_k^2}{q}(1 + \log t)^2 \right). \quad (\text{GRA18})$$

In particular the reference exponent has not been presumed larger than the arithmetic error on the whole of the full-form range.

Now take integers $s = s_k \geq 1$ and suppose

$$t = s/q \rightarrow \infty, \quad t \log t = o(q/k), \quad H_k = \frac{q}{t \log t}. \quad (\text{GRA19})$$

We prove every needed uniformity directly. The exact formula for q gives $q \geq k^2$, $q \leq C_m k^3$, and $R_k^2/q \leq \delta^2 + \gamma^2$ for $k \geq 1$. Eventually $t \geq e$ and $t \log t \leq q/k$, hence $t \leq q/k \leq C_m k^2$. It follows that

$$\log t = O_m(\log k), \quad \log(q + s + q) = O_m(\log k), \quad (R_k^2/q)(1 + \log t)^2 = O_h((\log k)^2) = o_h(k).$$

Thus the error in (GRA18) is $O_h(k)$, while $H_k/k = q/(kt \log t) \rightarrow \infty$. Also

$$\frac{R_k \log t}{q} = \sqrt{\delta^2 + \gamma^2} \frac{kt \log t}{q} \frac{1}{t} \rightarrow 0,$$

so the narrower range lies inside the proved full-form range.

For every integer $j \in \{-1, 0, \dots, q\}$ put $\tau_j = (s + j)/q$. Uniformly in this entire set,

$$|\tau_j - t| \leq 1, \quad \tau_j/t = 1 + O(1/t), \quad \log \tau_j = \log t + O(1/t), \quad \frac{\tau_j \log \tau_j}{t \log t} = 1 + O(1/t).$$

For example the logarithm estimate follows by the mean value theorem on $[t - 1, t + 1]$ once $t \geq 2$. These bounds prove the two range conditions and the $O_h(k)$ error uniformly for all τ_j . The definition of the scalar limit $2v \log v \psi(v) \rightarrow 1$ implies uniformity on all $v \geq t - 1$: the supremum of its absolute error on that tail tends to zero. Consequently

$$\max_{-1 \leq j \leq q} \left| \frac{\log T_{s+j}}{H_k} - 1 \right| \rightarrow 0. \quad (\text{GRA20})$$

The same argument applied to the finite bounds in (GRA11) shows that their lower and upper endpoints are $H_k(1 + o_h(1))$ uniformly. In particular all the finite lower endpoints are eventually positive; positivity has been proved in this range, not inserted as a new asymptotic premise.

For $r \geq 1$ take logarithms of the exact adjacent identity:

$$2 \log \epsilon_{q+r-1} = \log \beta_{q+r} + \log T_r + \log(1 - T_r^{-1}) + \log(1 - T_{r-1}^{-1}). \quad (\text{GRA21})$$

For $r = s, \dots, s + q$, (GRA17) and the preceding degree bounds give $\log \beta_{q+r} = O_h(k)$. If $M_k = \min_{s-1 \leq r \leq s+q} \log T_r$, then $M_k/H_k \rightarrow 1$ and

$$0 \leq -\log(1 - T_r^{-1}) \leq \frac{e^{-M_k}}{1 - e^{-M_k}} \rightarrow 0$$

uniformly for both adjacent factors. Divide (GRA21) by H_k and use $k/H_k \rightarrow 0$ to obtain

$$\max_{s \leq r \leq s+q} \left| \frac{\log \epsilon_{q+r-1}}{H_k/2} - 1 \right| \rightarrow 0. \quad (\text{GRA22})$$

For a single sequence r this proves $\log \epsilon_{q+r-1} \sim q/(2(r/q) \log(r/q))$ under the same narrower range. Since $\log \mathcal{L}_{h,k} = \log(\delta/2) + \log q + \log(k+1) = O_h(\log k)$, it also proves the original benchmark comparison

$$\log \frac{\epsilon_{q+r-1}}{\delta q(k+1)/2} \sim \frac{q}{2(r/q) \log(r/q)} \rightarrow +\infty. \quad (\text{GRA23})$$

These statements do not extend the allowance asymptotic to degrees for which the arithmetic error in (GRA18) is comparable to its decreasing reference exponent.

4.5 Exact shifted two-window identities and finite enclosures

Put $a = q + s - 1$ and $b = a + q$, and give the rows $N = a + i$, $0 \leq i \leq q$, the weights $w_0 = w_q = 1$, $w_i = 2$ otherwise. Their sum is $2q$. Since $s \geq 1$, $a \geq q$, so every row has both adjacent factors. Define

$$\begin{aligned} \mathcal{B}_{k,s}^{\text{ar}} &= \log \frac{V_a V_{a+1}}{V_b V_{b+1}}, & \mathcal{W}_{k,s}^{\text{ar}} &= \log \frac{\omega_b^{\text{ar}} \omega_{b+1}^{\text{ar}}}{\omega_a^{\text{ar}} \omega_{a+1}^{\text{ar}}}, \\ \mathcal{P}_{k,s} &= - \sum_{i=0}^q w_i [\log(1 - d_{a+i}) + \log(1 - d_{a+i+1})], & \mathcal{J}_{k,s}^{\text{action}} &= 2 \sum_{i=0}^q w_i \log \epsilon_{a+i}. \end{aligned} \quad (\text{GRA24})$$

For any positive sequence z_n , expansion of the coefficients gives

$$\sum_{i=0}^q w_i (\log z_{a+i+1} - \log z_{a+i}) = \log z_b + \log z_{b+1} - \log z_a - \log z_{a+1}.$$

Applying this identity to ω_n^{ar} and V_n in (GRA21), with $r = s + i$ and $\log T_{s+i} = -\log d_{a+i+1}$, proves

$$J_{k,s}^{\text{action}} = \mathcal{B}_{k,s}^{\text{ar}} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s}, \quad \mathcal{B}_{k,s}^{\text{ar}} = \sum_{i=0}^q w_i \log T_{s+i}. \quad (\text{GRA25})$$

The four endpoints are thus exactly $a, a+1, a+q, a+q+1$.

To derive the penalty weights, let $f(x) = -\log(1-x)$ for $0 < x < 1$. The coefficient of $f(d_{a+j})$ is $w_j + w_{j-1}$, taking a missing w as zero. Since $q \geq 2$ for the original packet, the result is

$$\mathcal{P}_{k,s} = \sum_{j=0}^{q+1} c_j f(T_{s-1+j}^{-1}), \quad (c_0, \dots, c_{q+1}) = (1, 3, \underbrace{4, \dots, 4}_{q-2}, 3, 1), \quad \sum_j c_j = 4q. \quad (\text{GRA26})$$

The first coefficient is present because the first row is shifted beyond the exceptional row $q-1$.

For finite bounds suppose all the lower endpoints in (GRA11) on $s-1 \leq r \leq s+q$ are positive, and define $\ell = \min_r \underline{E}_r > 0$ and $u = \max_r \bar{E}_r$. The derivative of f is positive, while e^{-x} decreases, so

$$\sum_{j=0}^{q+1} c_j f(e^{-\bar{E}_{s-1+j}}) \leq \mathcal{P}_{k,s} \leq \sum_{j=0}^{q+1} c_j f(e^{-\underline{E}_{s-1+j}}). \quad (\text{GRA27})$$

The elementary inequalities $x \leq f(x) \leq x/(1-x)$, proved by integrating $1/(1-x)$ between zero and x , give the useful common bounds

$$4qe^{-u} \leq \mathcal{P}_{k,s} \leq \frac{4qe^{-\ell}}{1-e^{-\ell}}. \quad (\text{GRA28})$$

Each individual final term also gives $\mathcal{P}_{k,s} \geq e^{-\bar{E}_{s+q}}$.

Every row has a corresponding explicit finite interval. Put $d = q + r$ and

$$\begin{aligned} A_r^- &= \frac{1}{2} [\log(d(d-1/2)) - L_{k,d} + \underline{E}_r + \log(1 - e^{-\underline{E}_r}) + \log(1 - e^{-\underline{E}_{r-1}})], \\ A_r^+ &= \frac{1}{2} [\log(d(d-1/2)) + L_{k,d-1} + \bar{E}_r + \log(1 - e^{-\bar{E}_r}) + \log(1 - e^{-\bar{E}_{r-1}})]. \end{aligned}$$

Each displayed scalar function of E is increasing on $E > 0$. Equations (GRA17) and (GRA21) therefore prove

$$A_r^- \leq \log \epsilon_{q+r-1} \leq A_r^+, \quad \min_{s \leq r \leq s+q} A_r^- \leq \log \min_{a \leq N \leq b} \epsilon_N \leq \min_{s \leq r \leq s+q} A_r^+. \quad (\text{GRA29})$$

The two positive contraction factors are retained in both endpoints.

For the monic part define the literal Gamma window

$$\mathcal{W}_{k,s}^\sigma = \log \frac{\gamma_b \gamma_{b+1}}{\gamma_a \gamma_{a+1}} = \log \frac{(2b)!(2b+2)!}{4^{2q}(2a)!(2a+2)!}. \quad (\text{GRA30})$$

Write $v_n = \log(\omega_n^{\text{ar}}/\gamma_n)$. Equation (GRA9) places v_n in $[\log A_k - L_{k,n}, \log A_k]$. Applying these four intervals with their actual signs, so the four $\log A_k$ terms cancel, proves

$$-L_{k,b} - L_{k,b+1} \leq \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{W}_{k,s}^\sigma \leq L_{k,a} + L_{k,a+1}. \quad (\text{GRA31})$$

Thus the recurrence contribution has four endpoint errors, rather than one independent arithmetic error for every row. In particular, writing P_- and P_+ for the lower and upper quantities in (GRA27), all finite data combine to give

$$\begin{aligned} \sum_{i=0}^q w_i \underline{E}_{s+i} + \mathcal{W}_{k,s}^\sigma - L_{k,b} - L_{k,b+1} - P_+ &\leq J_{k,s}^{\text{action}}, \\ J_{k,s}^{\text{action}} &\leq \sum_{i=0}^q w_i \bar{E}_{s+i} + \mathcal{W}_{k,s}^\sigma + L_{k,a} + L_{k,a+1} - P_-. \end{aligned} \quad (\text{GRA32})$$

4.6 Complete asymptotics and the unchanged mixed receiver

Under (GRA19), (GRA22) directly implies

$$\log \min_{a \leq N \leq a+q} \epsilon_N \sim \frac{H_k}{2} = \frac{q}{2t \log t}. \quad (\text{GRA33})$$

Indeed every row logarithm divided by $H_k/2$ lies between $1 - o(1)$ and $1 + o(1)$, so its minimum does too. This argument does not identify the minimizing row with an endpoint or equate a minimum with a geometric mean.

For the penalty, (GRA20) implies that its minimum and maximum $\log T_r$ are both $H_k(1 + o(1))$. Apply the proof of (GRA28) using these exact minima and maxima to obtain

$$\log(4q) - \max_r \log T_r \leq \log \mathcal{P}_{k,s} \leq \log(4q) - \min_r \log T_r - \log(1 - e^{-\min_r \log T_r}).$$

Here $\log q = O_m(\log k) = o(H_k)$, and the final logarithm tends to zero. Dividing by H_k proves the full logarithmic rate

$$\log \mathcal{P}_{k,s} \sim -H_k = -\frac{q}{t \log t}. \quad (\text{GRA34})$$

All $4q$ coefficients have been retained in the upper estimate.

The separate terms of the action can also be evaluated. From (GRA20) and $\sum_i w_i = 2q$, $\mathcal{B}_{k,s}^{\text{ar}} \sim 2qH_k$. The exact Gamma recurrence gives

$$\mathcal{W}_{k,s}^\sigma = \sum_{i=0}^q w_i \log[(s + q + i)(s + q + i - 1/2)].$$

Since $s = qt$, every summand equals $2 \log s + O(1/t)$ uniformly, giving $\mathcal{W}_{k,s}^\sigma = 4q \log(qt) + O(q/t)$. More explicitly, $\sum_i w_i i = q^2$, and $0 \leq \log(1 + x) \leq x$ for $x \geq 0$ gives

$$0 \leq \mathcal{W}_{k,s}^\sigma - 4q \log s \leq (6q^2 - q)/s.$$

The four-endpoint estimate (GRA31) now yields

$$\begin{aligned} \mathcal{B}_{k,s}^{\text{ar}} &\sim 2qH_k, & \mathcal{W}_{k,s}^{\text{ar}} &= 4q \log(qt) + O_h(q/t + k), \\ \mathcal{W}_{k,s}^{\text{ar}} &= o(qH_k), & \mathcal{P}_{k,s} &= \exp[-H_k(1 + o(1))] = o(qH_k), \\ J_{k,s}^{\text{action}} &\sim 2qH_k = \frac{2q^2}{t \log t}. \end{aligned} \quad (\text{GRA35})$$

For the smallness of $\mathcal{W}$, divide its bound by qH_k and use $\log(qt) = O_m(\log k)$, $H_k/k \rightarrow \infty$, and $q \rightarrow \infty$. Equation (GRA25) then proves the last assertion independently of summing the row asymptotics.

It remains to specify exactly how this action reaches the existing mixed calculation. Keep its original surjection $\Lambda : E \twoheadrightarrow B$ and kernel inclusion $I : K \hookrightarrow E$, with the complete divided Taylor unit and original period branches. Explicitly, with the original period map Π and tensor invariant projector P_k , its image inclusion j satisfies $j\Lambda = P_k \Pi^{\otimes k} M_{v_h}^{\otimes k} \iota$, where $\iota[P] = [P(z_1 + \cdots + z_k)]$. For each original source metric G_N^v , put

$$H_{K,N}^v = I^* G_N^v I, \quad Q_{B,N}^v = (\Lambda(G_N^v)^{-1} \Lambda^*)^{-1}, \quad S_N^v = (G_N^v)^{-1} \Lambda^* Q_{B,N}^v.$$

Direct multiplication gives $\Lambda S_N^v = 1$ and $I^* G_N^v S_N^v = 0$. For a fixed section S_* , the difference $S_N^v - S_*$ has image in K , so the frame change from $[I, S_*]$ to $[I, S_N^v]$ is triangular with identity diagonal. Hence

$$\det H_{K,N}^v \det Q_{B,N}^v = |\det[I, S_*]|^2 \det G_N^v.$$

Taking the return at $a, a+1, b, b+1$ cancels all four frame constants. If $F_K^v, F_B^v, \mathcal{B}^v$ denote these same four-endpoint returns, the result is exactly $F_K^v + F_B^v = \mathcal{B}^v$. Empty determinant factors are one. For the original sources $v = \text{ar}, \sigma, (k)$ set

$$\mathcal{H} = \mathcal{B}^\sigma - \mathcal{B}^{(k)}, \quad \Delta^\sigma = \mathcal{B}^{\text{ar}} - \mathcal{B}^\sigma.$$

The complete shifted allocation and matching budget therefore obey

$$\begin{aligned} \Psi &= F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} = F_K^{(k)} + \mathcal{H} + \Delta^\sigma, \\ C^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s} = \mathcal{B}^{(k)} - F_K^{(k)} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s}, \\ \Psi + C^{\text{budget}} &= J_{k,s}^{\text{action}}. \end{aligned} \tag{GRA36}$$

This proves the exact map of the new window into the earlier complete action. The common allocation cancels with its matching budget. The new scalar profile consequently assigns no separate leading coefficient to the intrinsic kernel and boundary mixed terms or to their complementary CPQ returns. Those original quantities remain in (GRA15) and (GRA36), with their maps unchanged.

5 Exact propagation to the earlier receiving arguments

Moving-cutoff bounds on the original receiving spaces (GRB1–4). The complete moving-cutoff proofs appended below act on precisely the spaces in RGI1–3. In particular, if $p \in \mathbb{C}[y]_{\leq j}$ and $U_j P(y) = i^{-j} P(k/2 + iy)$, their arithmetic middle expression is $\|\chi U_j^{-1} p\|_{\text{ar}}^2$. The identity $U_{q+j} M_\chi = M_Q U_j$ proves this assertion, including the unit phase on every lower coefficient. No remainder fibre or arithmetic denominator changes.

At a fixed integer j , collect every *applicable* proved full-form comparison $\ell H_0 \leq H_Q \leq u H_0$ in the following finite set:

provider and its full stated guards	ℓ	u	
RGC5–7	$(1 - \rho_{q,j}^2)^q (1 - \eta_{q,j})$	$(1 + \rho_{q,j}^2)^q + \eta_{q,j}$	(GRB1)
PCRP1–8: $q \geq 2^{17} R_k, j \leq 2q$	$e^{-C_k^{\text{cmp}}}$	$e^{C_k^{\text{cmp}}}$	
moving cutoff: $j \geq 3$ and both parity guards	$e^{-C_\chi(k,j)} (1 - \eta_{k,j}^{\text{mov}})$	$1 + \eta_{k,j}^{\text{mov}}$	

Use a row only where its own proof applies; η^{mov} distinguishes the new quantity from RGC’s earlier inner error. If the set is nonempty, define $\ell_j^* = \max \ell$ and $u_j^* = \min u$. Then $\ell_j^* H_0 \leq H_Q \leq u_j^* H_0$ for every complex polynomial of degree at most j , including zero. Indeed each inequality holds on that same space, so one may take the maximum of its lower constants and the minimum of its upper constants. This argument also proves $\ell_j^* \leq u_j^*$ by evaluating any nonzero polynomial in the positive norm.

Write $\mathcal{L}_{k,n} = \log(A_k D_n / a_k)$. Every proved lower and upper reference endpoint $l_j^0 \leq \log(\nu_j^0 / \gamma_{q+j}) \leq u_j^0$ gives

$$l_j^0 - \mathcal{L}_{k,q+j} + \log \ell_j^* \leq \log T_j^{\text{ar}} \leq u_j^0 + \mathcal{L}_{k,q+j} + \log u_j^*. \quad (\text{GRB2})$$

To prove this, minimize the same full-form inequalities over the original affine monic fibre, then bound the unchanged denominator ω_{q+j} by AMT. For the new moving reference, the exact endpoints are $l_j^0 = Z_{q,j} + \log c_-$ and $u_j^0 = Z_{q,j} + \log c_+$. The all-degree lower endpoint $F_{q,j} + \log C_0$ remains available from RGC16. There is also an explicit upper reference endpoint at every $j \geq 0$, including the degrees 0 and 1 outside the moving parity theorem. Set $n = q + j$ and use the actual monic trial $p(y) = y^j$. The exact original moment generating function in GRE15 gives

$$\int y^{2n} d\sigma(y) = \sqrt{2\pi} (2n)! [z^{2n}] (\cos z)^{-1/2}, \quad u_j^{\text{trial}} = \log \left(4^n [z^{2n}] (\cos z)^{-1/2} \right).$$

Differentiation of the generating integral is justified by its exponential envelope on a neighbourhood of zero. This coefficient is positive because it equals the positive moment divided by $\sqrt{2\pi} (2n)!$. The trial has norm equal to that moment, so the monic minimum is no larger, proving $\log(\nu_j^0 / \gamma_n) \leq u_j^{\text{trial}}$ with the original mass and factorials. Thus this endpoint can always be used in GRB2 when one of its full-form rows is available. The moving reference endpoints themselves require $j \geq 2$, and the moving full-form row requires $j \geq 3$. Maximize the resulting lower endpoints also with the previously proved RGI5/UCT endpoints wherever their guards hold; minimize all eligible upper endpoints. Denote the results by L_j, U_j . Thus each new comparison improves or preserves every eligible earlier bound, with no extension of its original domain.

Whenever $0 < L_j \leq U_j$ and, for $j \geq 1$, $0 < L_{j-1} \leq U_{j-1}$, set $b_j^- = (q+j)(q+j - \frac{1}{2}) e^{-\mathcal{L}_{k,q+j}}$ and $b_j^+ = (q+j)(q+j - \frac{1}{2}) e^{\mathcal{L}_{k,q+j-1}}$. The exact adjacent identity now yields

$$\begin{aligned} b_j^- (e^{L_j} - 1) (1 - e^{-L_{j-1}}) &\leq \epsilon_{q+j-1}^2 \\ &\leq b_j^+ (e^{U_j} - 1) (1 - e^{-U_{j-1}}), \quad j \geq 1. \end{aligned} \quad (\text{GRB3})$$

For $j = 0$, omit precisely the preceding contraction on both sides. AMT at the two adjacent monic degrees proves the displayed bounds on β_{q+j} ; each remaining factor increases in its exponent. This proves GRB3 without discarding either contraction.

For any of the actual windows, let c_j be its already derived nonnegative penalty coefficients, over precisely the participating relation indices. Use these finite logarithmic endpoints when $0 < L_j \leq U_j$ at every participating index. The complete finite penalty then has the two-sided bound

$$\sum_j c_j [-\log(1 - e^{-U_j})] \leq \mathcal{P} \leq \sum_j c_j [-\log(1 - e^{-L_j})]. \quad (\text{GRB4})$$

This follows term by term from the decreasing function $x \mapsto -\log(1 - e^{-x})$, for $x > 0$. For the unshifted window retain the coefficients in RGI7 and their total $4r - 1$; for the shifted $q + 1$ rows below retain $1, 3, 4, \dots, 4, 3, 1$ and total $4q$. Inserting these endpoints into the exact action identity updates its finite enclosure. The original arithmetic V endpoints and the four AMT errors remain in place.

Propagation to the complete earlier band and its degree frontier (GRB5–8). The new moving-cutoff full-form and reference proofs establish, at the original fixed complete quartet and along $k = 4\ell + 1 \rightarrow \infty$, that

$$t_k = r_k/q \rightarrow \infty, \quad t_k \log t_k = o(q/k) \implies H_k := \frac{q}{t_k \log t_k} \gg k. \quad (\text{GRB5})$$

Here and throughout $q = [1 + k(m-1)](k+1)^2$, with the complete integer multiplicity m fixed. The root radius remains $R_k = k\sqrt{\delta^2 + \gamma^2}$. The result also propagates to the entire original, unshifted band:

$$\min_{0 \leq j \leq r_k} \log T_j^{\text{ar}} \sim H_k, \quad \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \sim \frac{1}{2} H_k. \quad (\text{GRB6})$$

In particular this strengthens the earlier growing-band lower estimates RGI12–14 while preserving their finite formulas.

Here is a proof of the uniform assertion, including the small-degree end. For $0 \leq j \leq \lfloor q/100 \rfloor$, the proved original interval bound UCT8 and its full arithmetic comparison give $\log T_j^{\text{ar}} \geq q/3 - O_h(k + \log q)$. Its root guard $q \geq 2^{17} R_k$ holds eventually because $q/R_k \rightarrow \infty$. On any fixed compact interval $1/200 \leq j/q \leq C$, the actual moving parameters α, β and their supports range over compact subsets of the positive quadrant. This follows from their explicit formulas and the continuous EIQ inverse moment parametrization. Thus b has a positive lower bound, $m(v^2 - u^2) \rightarrow \infty$ uniformly, $\eta^{\text{mov}} = e^{-q \log 8 + O_C(\log q)}$, and $C_\chi = O_{h,C}(R_k^2/q)$. The finite moving reference bounds give

$$\log T_j^{\text{ar}} = 2q\psi(j/q) + O_{h,C}(k + \log q + R_k^2/q).$$

The profile is positive at every finite positive argument: the exact UCT derivative $\psi'(t) = \log(\kappa/E) < 0$ and its limit zero at infinity prove this. Its compact minimum is consequently positive. Since $R_k^2/q = O_h(1)$, all these compact-degree exponents exceed $c_C q$ eventually, for some $c_C > 0$.

For rigor in passing to the whole expanding set, choose any sequence of indices $0 \leq j_k \leq r_k$. Every subsequence on which j_k/q is bounded has a further subsequence covered by the preceding small or compact bound. There $\log T_{j_k}^{\text{ar}}/H_k \rightarrow \infty$. Every unbounded subsequence has a further subsequence with $j_k/q \rightarrow \infty$. Its ratio satisfies the same narrow condition because $u \log u$ increases for $u \geq 1$ and $j_k \leq r_k$. The newly proved arithmetic estimate then gives

$$\log T_{j_k}^{\text{ar}} = (1 + o(1)) \frac{q}{(j_k/q) \log(j_k/q)} \geq (1 - o(1)) H_k.$$

If a uniform lower bound $(1 - o(1)) H_k$ failed, choosing a violating index at each such k and applying this subsequence argument would contradict one of its two conclusions. At $j = r_k$ the new asymptotic supplies the matching upper bound. This proves the first part of GRB6.

AMT gives uniformly over this band $\log \beta_{q+j} \geq -\mathcal{L}_{k,q+r_k} + \log(q(q-1/2)) = -O_h(k + \log(q + r_k))$. The narrow condition implies $r_k/q = O_h(k^2)$ eventually, so this cost is $O_h(k) = o(H_k)$. The

exact adjacent formula, with its exceptional $j = 0$ factor omitted, and the just proved uniform lower bound give $\log \epsilon_{q+j-1} \geq (1 - o(1))H_k/2$ uniformly. Indeed the two logarithms $\log(1 - e^{-\log T_j})$ and, when present, $\log(1 - e^{-\log T_{j-1}})$ tend uniformly to zero. The endpoint $j = r_k$ again supplies the upper bound for the minimum. This proves the second part of GRB6. As $\log(\delta q(k+1)/2) = O_h(\log k) = o(H_k)$, the minimum divided by the unchanged exterior benchmark tends to infinity.

The original unshifted penalty also has its exact exponential rate:

$$\log \mathcal{P}_{q,r_k} \sim -H_k. \quad (\text{GRB7})$$

For its upper bound, all $4r_k - 1$ original coefficient units multiply terms at most $e^{-(1-o(1))H_k}/(1 - e^{-(1-o(1))H_k})$. For its lower bound retain its last term, whose coefficient is one: $-\log(1 - T_{r_k}^{-1}) \geq T_{r_k}^{-1} = e^{-(1+o(1))H_k}$. Since $\log(4r_k) = O_h(\log k) = o(H_k)$, taking logarithms proves GRB7. These are exactly the unshifted endpoints and coefficients from RGI8–10, not those of the separately derived shifted window.

For a concrete improved frontier put $(d, A_m^{\deg}) = (2, 1)$ at $m = 1$, and $(3, m - 1)$ at fixed $m \geq 2$. Fix real ζ and $0 < \theta < d - 1$, or fix $\theta = d - 1$ and $\zeta > 1$. For the actual integer cutoff $r_k = \lfloor qk^\theta/(\log k)^\zeta \rfloor$, GRB6 gives

$$\log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \sim \frac{A_m^{\deg}}{2\theta} k^{d-\theta} (\log k)^{\zeta-1}. \quad (\text{GRB8})$$

To check every exponent and floor, $q \sim A_m^{\deg} k^d$ and $r_k/q = k^\theta (\log k)^{-\zeta} + O(1/q)$, so $\log(r_k/q) = \theta \log k - \zeta \log \log k + o(1)$. The ratio of $t_k \log t_k$ to q/k is therefore asymptotic to $(\theta/A_m^{\deg}) k^{\theta-d+1} (\log k)^{1-\zeta}$. It tends to zero in each displayed domain, which proves applicability of GRB5. Substitution into $H_k/2$ gives GRB8. In particular the pure powers now extend to every $\theta < 1$ at $m = 1$ and every $\theta < 2$ at fixed $m \geq 2$, with the stated logarithmic approach to the new endpoints. The case $\theta = d - 1, \zeta = 1$ has a nonzero limiting ratio in GRB5; this calculation therefore gives its exact comparison to the admissibility condition but no claimed asymptotic there. Every eligible earlier finite estimate remains in GRB1–4.

An explicit arithmetic coefficient at the logarithmic endpoint (GRB11–13). The exact constants in GRA6 also permit a positive allowance estimate at a specified part of the critical scale $\zeta = 1$ in GRB8. This conclusion is an enclosure with its retained arithmetic coefficient. Put

$$\mathfrak{s}_h = \frac{\pi}{2} + \log \frac{M_{\pi/2}}{\vartheta_h}, \quad C_h^{\deg} = \frac{A_m^{\deg}}{2(d-1)\mathfrak{s}_h}, \quad 0 < c_{\deg} < C_h^{\deg}, \quad r_k = \left\lfloor \frac{c_{\deg} q k^{d-1}}{\log k} \right\rfloor. \quad (\text{GRB11})$$

The coefficient $c_{\deg}$ is fixed independently of k ; the original centre remains $c = k/2$. All source quantities here are already defined: $\vartheta_h = \int_{-1}^1 w_h(y) dy$ and $M_{\pi/2} = \int e^{\pi y/2} w_h(y) dy$. Evenness and positivity of w_h give $M_{\pi/2} = \int \cosh(\pi y/2) w_h(y) dy \geq \int w_h \geq \vartheta_h > 0$. The proved tilted tail bound makes the first integral finite. Thus $\mathfrak{s}_h \geq \pi/2 > 0$ and the endpoint coefficient $C_h^{\deg}$ is defined and positive, derived from the actual arithmetic source comparison.

Set $B = A_m^{\deg}/(c_{\deg}(d-1))$, so $B > 2\mathfrak{s}_h$. The original full-band minimum satisfies the explicit bounds

$$\frac{B}{2} - \mathfrak{s}_h \leq \liminf_{k \rightarrow \infty} \frac{1}{k} \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \leq \limsup_{k \rightarrow \infty} \frac{1}{k} \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \leq \frac{B}{2} + \mathfrak{s}_h. \quad (\text{GRB12})$$

In particular this minimum still exceeds the original exterior benchmark by a ratio tending to infinity.

We prove the bounds with the original error rather than extending the little- o assertion GRB5 to its boundary. Here $t_k = r_k/q \sim c_{\deg} k^{d-1}/\log k$ and $\log t_k \sim (d-1) \log k$, so $q/(kt_k \log t_k) \rightarrow B$. This sequence satisfies the broad full-form condition $R_k \log t_k/q \rightarrow 0$. The finite moving reference theorem and GRF22, with $R_k^2/q = O_h(1)$, yield uniformly on its polynomial degree horizon

$$C_\chi(k, j) = o(k), \quad \log(q + j) = o(k), \quad -\log(1 - \eta(k, j)) = o(k)$$

for the growing-ratio part of that horizon. To see the uniform meaning, its explicit root error is bounded by $O_h((1 + \log t_k)^2) = o(k)$; the logarithmic prefactors in its inner error are $O_h(\log k)$. On compact positive ratio intervals the same conclusion follows from the actual compact endpoint bounds used in the proof of GRB6. The exact expression GRA6 gives

$$L_{k,q+j}/k = \mathfrak{s}_h + o(1) \quad \text{uniformly for } 0 \leq j \leq r_k + q.$$

The degree-dependent term is $\log D_{q+j} = O_h(\log k)$ and all remaining terms of GRA6 except $k\mathfrak{s}_h$ are $O_h(\log k)$, proving this last display directly.

Using the same bounded-ratio/unbounded-ratio subsequence argument as in GRB6, the reference signal has $\min_{j \geq q/100, j \leq r_k} 2q\psi(j/q)/k \rightarrow B$; at $j = r_k$ it is $B + o(1)$ and for unbounded ratios below it the function $u \log u$ gives the lower estimate. For bounded ratios its quotient by k tends to infinity. The original small-degree estimate also tends to infinity after division by k . Therefore the actual arithmetic transfer proves

$$\min_{0 \leq j \leq r_k} \log T_j^{\text{ar}}/k \geq B - \mathfrak{s}_h - o(1) > 0, \quad B - \mathfrak{s}_h - o(1) \leq \log T_{r_k}^{\text{ar}}/k \leq B + \mathfrak{s}_h + o(1).$$

AMT at adjacent degrees gives $-\mathfrak{s}_h - o(1) \leq \log \beta_{q+j}/k \leq \mathfrak{s}_h + o(1)$ uniformly. Both contraction logarithms tend uniformly to zero, since their exponents are at least a positive multiple of k ; the first row retains its single-factor exception. The lower part of the exact adjacent identity now gives the lower bound in GRB12. Its endpoint $j = r_k$, using the upper recurrence and upper scalar bounds, gives the upper bound for the minimum. The original benchmark has logarithm $O_h(\log k) = o(k)$, proving the stated comparison.

For the shifted band with $s = r_k$, $a = q + s - 1$ and $b = a + q$, the exact complete action admits a sharper coefficient enclosure than summing the separate row recurrence errors:

$$2(B - \mathfrak{s}_h) \leq \liminf_{k \rightarrow \infty} \frac{J_{k,s}^{\text{action}}}{qk} \leq \limsup_{k \rightarrow \infty} \frac{J_{k,s}^{\text{action}}}{qk} \leq 2(B + \mathfrak{s}_h). \quad (\text{GRB13})$$

Indeed $|(s+j)/q - t_k| \leq 1$ for $-1 \leq j \leq q$, so the reference signal divided by k tends uniformly to B . The scalar arithmetic exponents therefore lie uniformly between $k(B - \mathfrak{s}_h - o(1))$ and $k(B + \mathfrak{s}_h + o(1))$. Their weighted sum has total weight $2q$, giving the two main endpoints in GRB13. The full penalty is at most $4qe^{-k(B - \mathfrak{s}_h - o(1))}/(1 - e^{-k(B - \mathfrak{s}_h - o(1))})$ and hence is $o(qk)$. Finally the exact Gamma recurrence window is $O(q \log k)$, and GRA31 charges only its four endpoint errors, each $O_h(k)$. Thus the full arithmetic recurrence window is $o(qk)$. Inserting these three estimates in GRB9 proves GRB13. This uses the complete correlated action, with no change to the intrinsic mixed allocation or the complementary returns.

The moving-degree result in the complete action (GRB9–10). The new growing-ratio proof also gives an exact receiver at shifted rows. Keep q, χ, E , all three source forms, and every kernel/boundary map fixed as in this section. Choose an integer $s \geq 1$, put $a = q + s - 1$, $b = a + q$, and give rows $N = a + i$ the weights $w_0 = w_q = 1$ and $w_i = 2$ otherwise. Their relation index is exactly $r = s + i$ in RGI8. Every row now has the preceding contraction, because $a \geq q$. Expanding the weighted differences for any positive sequence z gives

$$\sum_{i=0}^q w_i (\log z_{a+i+1} - \log z_{a+i}) = \log z_b + \log z_{b+1} - \log z_a - \log z_{a+1}.$$

Applying this separately to the original V and ω in the exact adjacent identity proves

$$\begin{aligned} J_{k,s}^{\text{action}} &= \log \frac{V_a V_{a+1}}{V_b V_{b+1}} + \log \frac{\omega_b \omega_{b+1}}{\omega_a \omega_{a+1}} - \sum_{j=0}^{q+1} c_j [-\log(1 - T_{s-1+j}^{-1})], \\ J_{k,s}^{\text{action}} &= 2 \sum_{i=0}^q w_i \log \epsilon_{a+i}, \quad (c_0, \dots, c_{q+1}) = (1, 3, \underbrace{4, \dots, 4}_{q-2}, 3, 1). \end{aligned} \quad (\text{GRB9})$$

Indeed $c_j = w_j + w_{j-1}$ with absent weights zero, so their total is $2 \sum_i w_i = 4q$. These four endpoints and every coefficient are derived from the original maps. The full finite scalar enclosures, four AMT endpoint errors and their proof are GRA27–32; GRB1–4 intersects them with all eligible earlier bounds. For $t = s/q \rightarrow \infty$ and $t \log t = o(q/k)$ the complete proof GRA19–35 consequently gives

$$\log \min_{a \leq N \leq b} \epsilon_N \sim \frac{q}{2t \log t}, \quad \log \mathcal{P}_{k,s} \sim -\frac{q}{t \log t}, \quad J_{k,s}^{\text{action}} \sim \frac{2q^2}{t \log t}. \quad (\text{GRB10})$$

The proof controls the full adjacent index interval $s - 1, \dots, s + q$; its error is uniform on that interval, as shown there before summing.

The exact transfer to the mixed spaces retains $I : K \hookrightarrow E$ and $\Lambda : E \twoheadrightarrow B$. For each unchanged positive metric G_N^v its minimum section is $S_N^v = (G_N^v)^{-1} \Lambda^* (\Lambda (G_N^v)^{-1} \Lambda^*)^{-1}$. It satisfies $\Lambda S_N^v = 1$ and $I^* G_N^v S_N^v = 0$ by direct multiplication. Relative to any fixed section S_* , the frame change from $[I, S_*]$ to $[I, S_N^v]$ has identity diagonal because $\Lambda(S_N^v - S_*) = 0$. Its determinant is one. Block orthogonality therefore gives

$$\det(I^* G_N^v I) \det(\Lambda (G_N^v)^{-1} \Lambda^*)^{-1} = |\det[I, S_*]|^2 \det G_N^v.$$

The four-endpoint return cancels that same frame constant and proves $F_K^v + F_B^v = \mathcal{B}^v$. Thus the complete allocation and its matching budget in GRA36 still sum to GRB9: the occurrence of $F_K^{(k)}$ in one has its opposite occurrence in the other. Every intrinsic mixed profile and complementary CPQ return retains its original map and coefficient; the new scalar estimates do not assign those separate coefficients.

A Exact equilibrium for the square-root logarithmic potential

Throughout this section $\alpha > 0$ and $\beta > 0$. On $(0, \infty)$ set

$$V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x, \quad \mathcal{E}_{\alpha,\beta}(\mu) = \int V_{\alpha,\beta}(x) d\mu(x) - \iint \log|x-y| d\mu(x)d\mu(y). \quad (\text{EIQ1})$$

The measures considered initially are probability measures for which $\int V_{\alpha,\beta} d\mu$ is finite. Because this potential is bounded below and tends to $+\infty$ at both ends, this implies $\int(\sqrt{x} + |\log x|) d\mu < \infty$. The positive part of $\log|x-y|$ is then integrable, by $\log^+|x-y| \leq \log(1+x) + \log(1+y)$. Thus the energy is well defined with value in $\mathbb{R} \cup \{+\infty\}$. All comparisons below also apply to the usual extended energy defined by integrating the kernel $\frac{1}{2}V_{\alpha,\beta}(x) + \frac{1}{2}V_{\alpha,\beta}(y) - \log|x-y|$: the kernel is bounded below, and finite energy for this kernel implies the same moment conditions. To see the latter assertion without cancellation, put $g(x) = \frac{1}{2}V_{\alpha,\beta}(x) - \log(1+x)$. The kernel is at least $g(x) + g(y)$; g is bounded below and controls positive constant multiples of both $\sqrt{x}$ near infinity and $|\log x|$ near zero.

A.1 Existence and uniqueness of the two endpoints

For $0 < k < 1$ define the complete elliptic integrals by their actual real integrals

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \theta} d\theta. \quad (\text{EIQ2})$$

There is exactly one $k \in (0, 1)$ satisfying

$$\frac{\sqrt{1-k^2}K(k)}{E(k)} = \frac{\alpha}{\alpha+2}. \quad (\text{EIQ3})$$

For this k put

$$r = \sqrt{1-k^2}, \quad v = \frac{(\alpha+2)\pi}{\beta E(k)}, \quad u = rv, \quad a = u^2, \quad b = v^2. \quad (\text{EIQ4})$$

In particular $0 < a < b < \infty$, and the endpoint equations, with all constants retained, are

$$uK(k) = \frac{\alpha\pi}{\beta}, \quad vE(k) = \frac{(\alpha+2)\pi}{\beta}, \quad k^2 = 1 - \frac{u^2}{v^2}. \quad (\text{EIQ5})$$

Here is a proof of the assertion of uniqueness. Differentiation of the integrals in (EIQ2), and integration of the derivative of $\sin \theta \cos \theta / \sqrt{1-k^2 \sin^2 \theta}$ between its two zero endpoint values, give

$$E'(k) = \frac{E(k) - K(k)}{k}, \quad K'(k) = \frac{E(k)}{k(1-k^2)} - \frac{K(k)}{k}.$$

For completeness the second identity is equivalent, after differentiation under the integral sign, to

$$\int_0^{\pi/2} \left\{ \frac{k^2 \sin^2 \theta}{(1-k^2 \sin^2 \theta)^{3/2}} - \frac{1-k^2 \sin^2 \theta}{(1-k^2)\sqrt{1-k^2 \sin^2 \theta}} + \frac{1}{\sqrt{1-k^2 \sin^2 \theta}} \right\} d\theta = 0;$$

the expression in braces is $-k^2(1-k^2)^{-1} d(\sin \theta \cos \theta / \sqrt{1-k^2 \sin^2 \theta})/d\theta$. With $t = E(k)/K(k)$, the integral definitions show strictly $r^2 < t < 1$. Consequently, if $f(k) = rK(k)/E(k)$, then

$$\frac{kf'(k)}{f(k)} = \frac{t}{r^2} + \frac{1}{t} - 1 - \frac{1}{r^2} = \frac{(t-1)(t-r^2)}{r^2 t} < 0. \quad (\text{EIQ6})$$

At $k = 0$ the continuous limiting value is $f(0) = 1$. As $k \uparrow 1$, the integrand of $rK(k)$ tends to zero at every $\theta < \pi/2$ and is bounded by 1, while $E(k) \rightarrow 1$ by dominated convergence. Hence

$f(k) \rightarrow 0$. This proves (EIQ3), including existence. It also proves smooth dependence of k, u, v, a, b on (α, β) in $(0, \infty)^2$, by the nonzero derivative in (EIQ6).

Let $m = (a + b)/2$, $R_s = \sqrt{(a + s)(b + s)}$ for $s \geq 0$, and $R_0 = uv$. Substitution $t = a \cos^2 \theta + b \sin^2 \theta$ gives, directly from (EIQ5),

$$\int_a^b \frac{V'_{\alpha, \beta}(t)}{\sqrt{(b-t)(t-a)}} dt = 0, \quad \int_a^b \frac{tV'_{\alpha, \beta}(t)}{\sqrt{(b-t)(t-a)}} dt = 2\pi. \quad (\text{EIQ7})$$

Indeed the three integrals with numerators $t^{-1/2}, t^{-1}, t^{1/2}$ are respectively $2K(k)/v$, $\pi/(uv)$, and $2vE(k)$, and $V'(t) = \beta/(2\sqrt{t}) - \alpha/t$.

A.2 A positive density and its exact Cauchy transform

The elementary Stieltjes integral, obtained by $s = xz^2$, is

$$\int_0^\infty \frac{ds}{\sqrt{s}(x+s)} = \frac{\pi}{\sqrt{x}} \quad (x > 0). \quad (\text{EIQ8})$$

Using this identity in (EIQ7) and applying Tonelli to its positive constituents gives

$$\frac{\alpha}{R_0} = \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}R_s}, \quad \int_0^\infty \frac{1 - s/R_s}{\sqrt{s}} ds = 2vE(k). \quad (\text{EIQ9})$$

Both integrals converge; their integrands are $O(s^{-1/2})$ at zero and $O(s^{-3/2})$ at infinity. The second formula also follows by averaging $\sqrt{t} = \pi^{-1} \int_0^\infty s^{-1/2} t/(t+s) ds$ against the arcsine probability measure on $[a, b]$.

Define for every $x > 0$

$$\begin{aligned} h(x) &= \frac{\alpha}{xR_0} - \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}(x+s)R_s} \\ &= \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x+s)R_s} ds > 0. \end{aligned} \quad (\text{EIQ10})$$

The second equality is exactly (EIQ9), with no change of the potential. All integrals in (EIQ10) converge absolutely and locally uniformly in $x > 0$. The candidate density on the original positive half-line is

$$\rho_{\alpha, \beta}(x) = \begin{cases} \frac{\sqrt{(b-x)(x-a)}}{2\pi} h(x), & a < x < b, \\ 0, & x \notin (a, b). \end{cases} \quad (\text{EIQ11})$$

It is strictly positive on (a, b) and vanishes continuously with the square-root factor at each endpoint. Positivity here follows from the last integral in (EIQ10), not from a one-interval ansatz.

The following elementary transform computes its mass and resolvent. Choose

$$R(z) = \sqrt{(z-a)(z-b)}$$

on $\mathbb{C} \setminus [a, b]$ with $R(z)/z \rightarrow 1$ at infinity. Thus $R(x) > 0$ for $x > b$, $R(x) < 0$ for $x < a$, and $R(x + i0) = i\sqrt{(b-x)(x-a)}$ on (a, b) . Direct substitution $t = m + (b-a)\cos\theta/2$, followed by $w = \tan(\theta/2)$, proves

$$\frac{1}{\pi} \int_a^b \frac{dt}{(z-t)\sqrt{(b-t)(t-a)}} = \frac{1}{R(z)}.$$

For real $z > b$, the substitution reduces the integral to $\pi^{-1} \int_0^\infty 2dw/[(z-b) + (z-a)w^2]$; the identity elsewhere follows by analytic continuation. Polynomial division then gives

$$\frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{z-t} dt = z - m - R(z), \quad \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{t+s} dt = m + s - R_s. \quad (\text{EIQ12})$$

For example the first identity follows from $(b-t)(t-a) = -(t-z)^2 + 2(m-z)(t-z) - (z-a)(z-b)$, the preceding arcsine transform, and the arcsine first moment m . Partial fractions in these two identities yield

$$H_s(z) := \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{(t+s)(z-t)} dt = 1 - \frac{R_s + R(z)}{z+s}. \quad (\text{EIQ13})$$

The formula at $z = -s$ is interpreted by its removable limit.

The mass of (EIQ11), computed using (EIQ12), is

$$\begin{aligned} \int_a^b \rho_{\alpha,\beta}(t) dt &= \frac{\alpha}{2R_0}(m - R_0) - \frac{\beta}{4\pi} \int_0^\infty \frac{(m+s)/R_s - 1}{\sqrt{s}} ds \\ &= -\frac{\alpha}{2} + \frac{\beta}{4\pi} \int_0^\infty \frac{1-s/R_s}{\sqrt{s}} ds = -\frac{\alpha}{2} + \frac{\beta v E(k)}{2\pi} = 1. \end{aligned} \quad (\text{EIQ14})$$

In particular this is a probability density.

Define its Cauchy transform with the explicit sign convention $G(z) = \int_a^b \rho_{\alpha,\beta}(t)/(z-t) dt$. For z outside both $[a, b]$ and $(-\infty, 0]$, formulas (EIQ10) and (EIQ13) give

$$\begin{aligned} G(z) &= \frac{\alpha}{2R_0} H_0(z) - \frac{\beta}{4\pi} \int_0^\infty \frac{H_s(z)}{\sqrt{s} R_s} ds \\ &= -\frac{\alpha}{2z} + \frac{\beta}{4\sqrt{z}} - \frac{R(z)h(z)}{2} = \frac{V'_{\alpha,\beta}(z) - R(z)h(z)}{2}. \end{aligned} \quad (\text{EIQ15})$$

Here the positive square root is continued with its negative-real-axis cut, and $h(z)$ is the first integral formula of (EIQ10). Every integral exchange is absolutely convergent, uniformly on compact subsets of this domain: near zero the auxiliary integrand is $O(s^{-1/2})$, and near infinity the original integral defining H_s is $O(s^{-1})$. The constant terms on the second line cancel by (EIQ9), and (EIQ8) supplies the term $\beta/(4\sqrt{z})$.

A.3 The Euler equality, its strict exterior inequality, and uniqueness

Set

$$U(x) = V_{\alpha,\beta}(x) - 2 \int_a^b \log|x-t| \rho_{\alpha,\beta}(t) dt \quad (x > 0). \quad (\text{EIQ16})$$

This function is continuous on $(0, \infty)$. The integral is continuous because the density is bounded, compactly supported, and the integral of $|\log|x-t||$ over an interval of length ε tends to zero uniformly in its centre. In the interior of $[a, b]$, differentiation of the logarithmic potential is the principal-value integral $\text{PV} \int \rho(t)/(x-t) dt$. This follows by subtracting $\rho(x)$ on a symmetric interval about x ; the remaining numerator is $O(|t-x|)$ because the density is continuously differentiable locally in the interior. The real part of (EIQ15) on the upper edge of the interval is precisely this principal value. Indeed subtracting $\rho(x)$ in $\int \rho(t)/(x+i\varepsilon-t) dt$ gives the same limit, while the constant term has real part equal to its elementary principal value. Since $R(x+i0)$ is purely imaginary and $h(x)$ is real, (EIQ15) proves $U'(x) = 0$ on (a, b) . There is therefore a real constant ℓ such that

$$U(x) = \ell \quad (a \leq x \leq b). \quad (\text{EIQ17})$$

For positive x outside this interval, (EIQ15) instead gives the exact real derivative

$$U'(x) = R(x)h(x) \begin{cases} < 0, & 0 < x < a, \\ > 0, & x > b. \end{cases} \quad (\text{EIQ18})$$

Combining (EIQ17), continuity, and these signs proves

$$U(x) > \ell \quad \text{for } x \in (0, a) \cup (b, \infty). \quad (\text{EIQ19})$$

This verifies the entire exterior inequality on the original domain, including the interval next to zero.

Here is a complete positivity argument for the energy of the difference of two measures. If ν is a finite signed real measure of total mass zero for which $|\log|x-y||$ is integrable against $|\nu| \otimes |\nu|$, the scalar identity

$$-\log d = \frac{1}{2} \int_0^\infty \frac{e^{-td^2} - e^{-t}}{t} dt \quad (d > 0) \quad (\text{EIQ20})$$

follows by differentiating in d and evaluating at $d = 1$. One half of the integral over any finite subinterval of $(0, \infty)$ has the same sign as $-\log d$ and absolute value at most $|\log d|$. Consequently dominated convergence and the zero total mass give

$$-\iint \log|x-y| d\nu(x)d\nu(y) = \frac{1}{2} \int_0^\infty \frac{1}{t} \iint e^{-t(x-y)^2} d\nu(x)d\nu(y) dt \geq 0. \quad (\text{EIQ21})$$

The nonnegative sign is proved, rather than assumed, by the Gaussian Fourier identity

$$\iint e^{-t(x-y)^2} d\nu(x)d\nu(y) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-\xi^2/(4t)} |\widehat{\nu}(\xi)|^2 d\xi, \quad \widehat{\nu}(\xi) = \int e^{i\xi x} d\nu(x). \quad (\text{EIQ22})$$

The scalar Gaussian Fourier identity follows by completing the square in the Gaussian integral (or by differentiating its Fourier transform and its value at zero); its integration against the finite measures is justified by absolute integrability. Equality in (EIQ21) forces $\widehat{\nu} = 0$: the right side of (EIQ22) is strictly positive for every $t > 0$ if this continuous Fourier transform has a nonzero value. Finally $\widehat{\nu} = 0$ implies $\nu = 0$ without an extra uniqueness assumption. Convolve ν with a centred Gaussian approximate identity. Its value at every point is zero by that same Fourier integral. For every compactly supported continuous test function, its convolution with these Gaussians converges uniformly to the test function; integration against ν then proves that ν annihilates every such function.

Let μ be any probability measure of finite energy in (EIQ1), and write $\mu_*(dx) = \rho_{\alpha,\beta}(x) dx$ and $\nu = \mu - \mu_*$. The integrability needed for (EIQ21) holds. The $\mu \otimes \mu$ term has it by finite energy and the already established logarithmic moment; the $\mu_* \otimes \mu_*$ term has it by bounded compact support; and the cross term has it because the negative logarithmic singularity integrated against the bounded density is uniformly bounded, while the positive part is controlled by $\log(1+x) + \log(1+y)$. Expanding the energy therefore yields exactly

$$\mathcal{E}_{\alpha,\beta}(\mu) - \mathcal{E}_{\alpha,\beta}(\mu_*) = \int (U(x) - \ell) d\mu(x) - \iint \log|x-y| d\nu(x)d\nu(y) \geq 0. \quad (\text{EIQ23})$$

The first term is nonnegative by (EIQ17)–(EIQ19), and the second is strictly positive unless $\mu = \mu_*$. Infinite energy competitors cannot improve this finite value. Thus (EIQ11) is the unique global minimizing probability measure, with exactly the support $[a, b]$.

A.4 An explicit endpoint integral for the logarithmic moment

Retain m, R_s, u, v as defined above, and define

$$C = \frac{(u+v)^2}{4}, \quad \eta = \frac{v-u}{v+u}, \quad \zeta_s = \frac{b-a}{2(m+s+R_s)} \quad (s \geq 0). \quad (\text{EIQ24})$$

Thus $0 < \zeta_s \leq \eta < 1$ and $\zeta_0 = \eta$. For every $s \geq 0$ put

$$L_s = (m+s-R_s) \log C - (m-R_0) - 2R_s \log(1-\eta\zeta_s). \quad (\text{EIQ25})$$

Then the logarithmic moment of the equilibrium measure is exactly

$$\mathcal{L}(\alpha, \beta) := \int_a^b \log x \rho_{\alpha,\beta}(x) dx = \frac{\beta}{4\pi} \int_0^\infty \frac{L_0 - L_s}{\sqrt{s} R_s} ds. \quad (\text{EIQ26})$$

This is a convergent one-dimensional endpoint integral of explicitly given real elementary functions; its only endpoints are those specified uniquely in (EIQ3)–(EIQ5).

Here is a direct derivation, including the constants in (EIQ25). Set $h_0 = (b-a)/2$ and substitute $t = m + h_0 \cos \theta$ in arcsine integrals. The identities

$$t = C(1 + 2\eta \cos \theta + \eta^2), \quad t + s = \frac{m + s + R_s}{2}(1 + 2\zeta_s \cos \theta + \zeta_s^2)$$

give the absolutely convergent Fourier series

$$\log t = \log C + 2 \sum_{j \geq 1} \frac{(-1)^{j+1} \eta^j}{j} \cos(j\theta), \quad \frac{1}{t+s} = \frac{1}{R_s} \left(1 + 2 \sum_{j \geq 1} (-\zeta_s)^j \cos(j\theta) \right).$$

Orthogonality on $[0, \pi]$ gives

$$\begin{aligned} \frac{1}{\pi} \int_a^b \frac{\log t}{\sqrt{(b-t)(t-a)}} dt &= \log C, \\ \frac{1}{\pi} \int_a^b \frac{t \log t}{\sqrt{(b-t)(t-a)}} dt &= m \log C + (m - R_0), \\ \frac{1}{\pi} \int_a^b \frac{\log t}{(t+s)\sqrt{(b-t)(t-a)}} dt &= \frac{\log C + 2 \log(1 - \eta \zeta_s)}{R_s}. \end{aligned} \tag{EIQ27}$$

The second identity uses $h_0 \eta = m - R_0$. Polynomial division

$$\frac{(b-t)(t-a)}{t+s} = -t + 2m + s - \frac{R_s^2}{t+s}$$

now proves, exactly,

$$L_s = \frac{1}{\pi} \int_a^b \frac{\log t \sqrt{(b-t)(t-a)}}{t+s} dt. \tag{EIQ28}$$

Using the positive integral expression for h in (EIQ10), and the identity $1/[t(t+s)] = (1/t - 1/(t+s))/s$, yields (EIQ26). The exchanges are absolutely convergent: $\log t$ is bounded on $[a, b]$, $L_0 - L_s = O(s)$ at zero by (EIQ28), and $L_s = O(s^{-1})$ at infinity. Thus the integrand in (EIQ26) is $O(s^{1/2})$ at zero and $O(s^{-3/2})$ at infinity.

A.5 Parameter continuity and the variational derivative

The density and its logarithmic moment depend smoothly on $(\alpha, \beta) \in (0, \infty)^2$ in the following precise sense. On a compact parameter neighbourhood, (EIQ3)–(EIQ6) put a, b in a fixed compact subset of $(0, \infty)$, with $b - a$ bounded below by a positive number. The integral for h and each of its parameter and x derivatives converge uniformly for x in this fixed compact set: their absolute integrands are bounded by constants times $s^{1/2}$ near zero and $s^{-3/2}$ near infinity. Parameter derivatives of the endpoints are bounded on a smaller compact neighbourhood. Under the fixed-interval coordinate $x = a + (b-a)y$, $0 \leq y \leq 1$, the measure is exactly

$$\rho_{\alpha, \beta}(x) dx = \frac{(b-a)^2}{2\pi} \sqrt{y(1-y)} h(a + (b-a)y) dy. \tag{EIQ29}$$

All parameter derivatives of its coefficient are bounded by a constant times $\sqrt{y(1-y)}$. The same is true after multiplication by $\log(a + (b-a)y)$ or by $\sqrt{a + (b-a)y}$. Differentiation under this integral proves smoothness, in particular continuity, of both moments on the entire positive parameter quadrant.

Define $E_{\min}(\alpha, \beta) = \mathcal{E}_{\alpha, \beta}(\mu_{\alpha, \beta})$, where the measure is the unique minimizer proved above. Testing the minimizer for a parameter pair against the neighbouring potential, and then reversing the two tests, gives for $h > 0$

$$-\mathcal{L}(\alpha + h, \beta) \leq \frac{E_{\min}(\alpha + h, \beta) - E_{\min}(\alpha, \beta)}{h} \leq -\mathcal{L}(\alpha, \beta).$$

The corresponding inequalities for $h < 0$ and the proved continuity therefore give

$$\partial_{\alpha} E_{\min}(\alpha, \beta) = -\mathcal{L}(\alpha, \beta), \quad \partial_{\beta} E_{\min}(\alpha, \beta) = \int \sqrt{x} d\mu_{\alpha, \beta}(x). \quad (\text{EIQ30})$$

No unproved differentiability of a minimizing selection is being used. For the opposite convention $F(\alpha, \beta) = \sup_{\mu} \{ \iint \log |x - y| d\mu(x) d\mu(y) - \int V_{\alpha, \beta} d\mu \} = -E_{\min}(\alpha, \beta)$, the exact derivative is therefore $\partial_{\alpha} F(\alpha, \beta) = \mathcal{L}(\alpha, \beta)$.

The second moment in (EIQ30) has an exact value. Push the minimizing measure forward by the explicit dilation $x \mapsto c^2 x$, $c > 0$. Writing $M = \int \sqrt{x} d\mu_{\alpha, \beta}(x)$, its energy minus the energy at $c = 1$ is precisely $\beta(c - 1)M - 2(\alpha + 1) \log c$. Its derivative at the minimizing value $c = 1$ is zero. Hence

$$\int \sqrt{x} d\mu_{\alpha, \beta}(x) = \frac{2(\alpha + 1)}{\beta}. \quad (\text{EIQ31})$$

The same explicit dilation, or (EIQ3)–(EIQ5), proves that the map $x \mapsto (\pi/\beta)^2 x$ pushes $\mu_{\alpha, \pi}$ to $\mu_{\alpha, \beta}$, including the density and its two endpoints. Consequently

$$\mathcal{L}(\alpha, \beta) = \mathcal{L}(\alpha, \pi) + 2 \log(\pi/\beta). \quad (\text{EIQ32})$$

For the particular original limiting potential $V(x) = \pi\sqrt{x} - 2 \log x$, the substitutions are exactly $\alpha = 2$, $\beta = \pi$. Equations (EIQ5), (EIQ10), and (EIQ26) become

$$uK(k) = 2, \quad vE(k) = 4, \quad h(x) = \frac{1}{2x} \int_0^{\infty} \frac{\sqrt{s}}{(x+s)R_s} ds, \quad \mathcal{L}(2, \pi) = \frac{1}{4} \int_0^{\infty} \frac{L_0 - L_s}{\sqrt{s}R_s} ds. \quad (\text{EIQ33})$$

In particular the preceding proof includes positivity, unit mass, the full Euler inequality, the unique global minimizer, and the parameter continuity needed at these actual parameters.

B The exact closed equilibrium coefficient

This section retains the potential, measure, and energy of EIQ1–33:

$$\begin{aligned} V_{\alpha,\beta}(x) &= \beta\sqrt{x} - \alpha \log x, \quad \alpha > 0, \quad \beta > 0, \quad x > 0, \\ \mu_{\alpha,\beta}(dx) &= \rho_{\alpha,\beta}(x) dx, \quad F(\alpha, \beta) = \iint \log|x-y| d\mu_{\alpha,\beta}(x) d\mu_{\alpha,\beta}(y) - \int V_{\alpha,\beta} d\mu_{\alpha,\beta}, \\ \mathcal{L}(\alpha, \beta) &= \int \log x d\mu_{\alpha,\beta}(x). \end{aligned} \quad (\text{ECL1})$$

The preceding complete EIQ proof supplies the unique minimizing probability measure, its positive density and exact support $[a, b] = [u^2, v^2]$, and its Euler equality. In particular, no assertion in this section changes the minimization problem or chooses another measure. The endpoints are exactly

$$uK(k) = \frac{\alpha\pi}{\beta}, \quad vE(k) = \frac{(\alpha+2)\pi}{\beta}, \quad k^2 = 1 - \frac{u^2}{v^2}, \quad 0 < u < v. \quad (\text{ECL2})$$

Here K and E have the modulus convention of EIQ2, not the elliptic-parameter convention. Define the original endpoint quantities

$$w = \frac{u+v}{2}, \quad z = uv, \quad c_{\text{cap}} = \frac{v^2 - u^2}{4}, \quad d\nu(x) = \frac{dx}{\pi\sqrt{(b-x)(x-a)}} \quad (a < x < b). \quad (\text{ECL3})$$

The letter z in ECL3 is a positive scalar. A Cauchy-transform variable below is denoted by λ . The measure ν has mass one by $x = (a+b)/2 + (b-a)\cos\theta/2$, $0 \leq \theta \leq \pi$.

B.1 Arcsine identities with their original constants

Put $m = (a+b)/2$, $d = (b-a)/2$, and $\eta = (v-u)/(v+u)$. Then $0 < \eta < 1$ and

$$x = m + d \cos \theta = w^2(1 + 2\eta \cos \theta + \eta^2), \quad z = w^2(1 - \eta^2).$$

The absolutely uniformly convergent series

$$\begin{aligned} \log x &= 2 \log w + 2 \sum_{n \geq 1} \frac{(-1)^{n+1} \eta^n}{n} \cos(n\theta), \\ \frac{1}{x} &= \frac{1}{z} \left(1 + 2 \sum_{n \geq 1} (-\eta)^n \cos(n\theta) \right) \end{aligned}$$

follow from the geometric series and the integral of that series. Termwise integration, and orthogonality of the cosines, prove

$$\begin{aligned} \int \log x d\nu(x) &= 2 \log w, \quad \int \frac{d\nu(x)}{x} = \frac{1}{z}, \\ \int \frac{\log x}{x} d\nu(x) &= \frac{2 \log w + 2 \log(1 - \eta^2)}{z} = \frac{2 \log z - 2 \log w}{z}. \end{aligned} \quad (\text{ECL4})$$

In the product of the two Fourier series, the contribution of degree $n \geq 1$ to the constant coefficient is $-2\eta^{2n}/(nz)$, giving the displayed logarithm without an omitted constant.

The logarithmic potential of ν on its entire support is

$$\int_a^b \log|x-t| d\nu(x) = \log c_{\text{cap}}, \quad a \leq t \leq b. \quad (\text{ECL5})$$

Here is a direct proof, including the endpoint values. Write $t = m + d \cos \phi$ and use

$$|m + d \cos \theta - t| = \frac{d}{2} |e^{i\theta} - e^{i\phi}| |e^{i\theta} - e^{-i\phi}|.$$

The integrand depending on $\cos \theta$ has the same mean on $[0, \pi]$ as on $[0, 2\pi]$. For $0 < r < 1$, the power series of $\log |1 - re^{i\theta}|$ has zero circle mean. This mean tends to the mean at $r = 1$: near its only singular angle, $|1 - re^{i\theta}|^2 = (1 - r)^2 + 2r(1 - \cos \theta)$, so for $r \geq 1/2$ the absolute value of its negative logarithm is bounded by $C + |\log |\theta||$, and its positive logarithm is bounded by $\log 2$. These are integrable bounds. A rotation proves that each of the two circle means above is zero, for every ϕ , including $\phi = 0, \pi$. Hence their sum is $\log(d/2) = \log c_{\text{cap}}$, proving ECL5.

All logarithmic double integrals used below are absolutely integrable. In fact the absolute logarithm for two arcsine variables is bounded by a constant plus

$$\left| \log \left| \sin \frac{\theta - \phi}{2} \right| \right| + \left| \log \left| \sin \frac{\theta + \phi}{2} \right| \right|.$$

Each term is integrable on $[0, \pi]^2$; near a zero, use $\sin t \geq 2t/\pi$ for $0 \leq t \leq \pi/2$, and integrate $|\log t|$. Thus the same assertion holds for any two signed measures whose absolute values are bounded by constant multiples of ν . The EIQ density also has this domination, because $\rho(x)/[1/(\pi\sqrt{(b-x)(x-a)})]$ is bounded on $[a, b]$.

Let ℓ have the exact EIQ17 convention

$$V_{\alpha,\beta}(x) - 2 \int \log |x - t| d\mu_{\alpha,\beta}(t) = \ell \quad (a \leq x \leq b).$$

Integrating this equality against ν , using ECL4–5 and EIQ7, gives

$$\boxed{\ell = 2(\alpha + 2) - 2\alpha \log w - 2 \log c_{\text{cap}}.} \quad (\text{ECL6})$$

Indeed $\int \sqrt{x} d\nu(x) = 2vE(k)/\pi$, whose product with β is $2(\alpha + 2)$ by ECL2. Fubini is justified by the absolute integrability just proved and the unit mass of μ . Smoothness of ℓ now follows from the endpoint smoothness in EIQ6.

B.2 Differentiation with the moving endpoints retained

All derivatives in this subsection are with respect to α at fixed β . Put

$$p_\alpha(x) = \frac{h_{\alpha,\beta}(x)}{2\pi}, \quad D(x) = (b - x)(x - a).$$

In this subsection a dot denotes a parameter derivative, so that $\dot{p}(x)$ differentiates the function $p_\alpha(x)$ at fixed x . The function $h_{\alpha,\beta}$ is exactly EIQ10; the subscript on p_α labels the parameter and does not denote differentiation. EIQ10 and EIQ29 prove that p_α and its parameter and spatial derivatives are bounded on a fixed positive compact neighbourhood of the supports when parameters range in a sufficiently small compact neighbourhood.

The derivative of the probability measure, interpreted by integration against every continuously differentiable function on this neighbourhood, is the finite signed measure

$$\sigma(dx) = \left[\dot{p}(x) \sqrt{D(x)} + \frac{p_\alpha(x)}{2\sqrt{D(x)}} \{ \dot{b}(x - a) - \dot{a}(b - x) \} \right] \mathbf{1}_{(a,b)}(x) dx, \quad (\text{ECL7})$$

$$\frac{\partial}{\partial \alpha} \int f d\mu_{\alpha,\beta} = \int f d\sigma, \quad \sigma([a, b]) = 0.$$

This identity includes the moving support. To prove it without postulating endpoint terms, set $\delta = b - a$ and $x = a + \delta y$. The measure in the fixed variable y is

$$\delta^2 p_\alpha(a + \delta y) \sqrt{y(1 - y)} dy.$$

Its differentiated integral against f is

$$\int_0^1 \sqrt{y(1-y)} \left[2\delta\dot{\sigma} p_\alpha f + \delta^2(\dot{p} + p'_\alpha \dot{x})f + \delta^2 p_\alpha f' \dot{x} \right] dy, \quad \dot{x} = \dot{a} + \dot{\delta}y.$$

Replace $\delta f'$ in the last summand by $df(x)/dy$ and integrate by parts. Its boundary term is $[\delta p_\alpha(x)\dot{x}\sqrt{y(1-y)}f(x)]_0^1 = 0$. The remaining terms combine, on returning to x , to $\dot{p}\sqrt{D} + p_\alpha \dot{D}/(2\sqrt{D})$, where $\dot{D} = \dot{b}(x-a) - \dot{a}(b-x)$, exactly ECL7. This integration by parts is justified first on $[\varepsilon, 1-\varepsilon]$ and then by passage to the limit: the boundary expression tends to zero and the differentiated square-root factor is bounded by a constant times $[y(1-y)]^{-1/2}$, an integrable function. The differentiated fixed-interval integral is valid by the uniform bounds above. Taking $f = 1$ proves the zero total mass in ECL7. The displayed density also proves $|\sigma| \leq C\nu$. In particular no endpoint atom has appeared or been discarded.

For fixed $x \in (a, b)$, differentiating the Euler equality gives

$$-\log x - 2 \int \log|x-t| d\sigma(t) = \dot{\ell}. \quad (\text{ECL8})$$

For precision, choose a smooth function of t equal to one near x , with support in a common interior subinterval for nearby parameters. On this subinterval the density and its parameter derivative are uniformly bounded, so differentiation of its logarithmic integral follows by domination by $C|\log|x-t||$. The complementary integrand is a continuously differentiable test function, so ECL7 applies. This proves ECL8 despite the logarithmic singularity and the moving endpoints.

We next identify σ exactly:

$$\boxed{\frac{\partial}{\partial \alpha} \mu_{\alpha, \beta}(dx) = \sigma(dx) = \frac{1}{2} \left(1 - \frac{z}{x}\right) d\nu(x).} \quad (\text{ECL9})$$

The derivative has the test-function meaning proved in ECL7, which in particular applies to the logarithmic and square-root moments because $a > 0$.

Here is an explicit uniqueness argument establishing ECL9. Denote its proposed right side by σ_0 . By ECL4 it has zero total mass, and $|\sigma_0| \leq C\nu$. Retain the branch $R(\lambda) = \sqrt{(\lambda-a)(\lambda-b)}$ with $R(\lambda)/\lambda \rightarrow 1$ at infinity, exactly as in EIQ12. The arcsine transform proved there and partial fractions yield

$$\begin{aligned} \int \frac{d\sigma_0(t)}{\lambda-t} &= \frac{1}{2R(\lambda)} - \frac{z}{2\lambda} \left(\frac{1}{z} + \frac{1}{R(\lambda)} \right) \\ &= \frac{1-z/\lambda}{2R(\lambda)} - \frac{1}{2\lambda}. \end{aligned} \quad (\text{ECL10})$$

There is no pole at zero: $R(0) = -uv = -z$, so the numerator $(\lambda-z)/R(\lambda) - 1$ vanishes there. The integral itself is analytic in a neighbourhood of zero, as is also immediate from $t \geq a > 0$. For $a < x < b$, the real principal value of the arcsine transform is zero, because $R(x+i0)$ is purely imaginary. This limiting principal value also follows directly by subtracting the local value of the smooth interior density in its Cauchy integral. Consequently ECL10 gives

$$\text{PV} \int \frac{d\sigma_0(t)}{x-t} = -\frac{1}{2x}.$$

Differentiating the logarithmic potential on each interior subinterval is valid by that same local subtraction; the endpoint parts are integrable and are a positive distance away. It follows that $-\log x - 2 \int \log|x-t| d\sigma_0(t)$ is constant on (a, b) . Subtract this identity from ECL8. For $\sigma_1 = \sigma - \sigma_0$, the logarithmic potential $\int \log|x-t| d\sigma_1(t)$ is constant on (a, b) . Since σ_1 has zero mass and no endpoint atoms, and $|\sigma_1| \leq C\nu$, absolute integrability proved after ECL5 gives

$$-\iint \log|x-t| d\sigma_1(x) d\sigma_1(t) = 0.$$

The exact zero-mass energy argument EIQ20–22, including its Fourier-uniqueness proof, forces $\sigma_1 = 0$. This establishes the claimed derivative as a measure, with every endpoint contribution in ECL7 retained.

B.3 The moment and energy formulas

Integrate ECL8 against ν . By ECL5 and $\sigma([a, b]) = 0$, the iterated logarithmic term is zero. Therefore

$$\dot{\ell} = -2 \log w. \quad (\text{ECL11})$$

All exchanges again have the absolute integrability established after ECL5; the equality ECL8 holds ν -almost everywhere. Using ECL9 with $f(x) = \log x$, and then ECL4, proves

$$\mathcal{L}_\alpha = \frac{1}{2} \int \left(1 - \frac{z}{x}\right) \log x \, d\nu(x) = 2 \log w - \log z = \log(w^2/z). \quad (\text{ECL12})$$

Write $I = \iint \log |x - t| \, d\mu(x) d\mu(t)$, with the original unscaled logarithmic kernel. The EIQ31 dilation identity is $\beta \int \sqrt{x} \, d\mu = 2(\alpha + 1)$. Integration of the Euler equality against μ gives $\ell = \int V \, d\mu - 2I$. Combining this with $F = I - \int V \, d\mu$ yields the exact identity

$$\ell = -2F - 2(\alpha + 1) + \alpha \mathcal{L}. \quad (\text{ECL13})$$

EIQ30 proves $F_\alpha = \mathcal{L}$. Differentiating ECL13 at fixed β , with the differentiability just proved, gives $\dot{\ell} = -\mathcal{L} - 2 + \alpha \mathcal{L}_\alpha$. Substituting ECL11–12 now proves, without an integration constant or a limiting parameter choice,

$$\mathcal{L} = 2(\alpha + 1) \log w - \alpha \log z - 2, \quad F = -\frac{\ell}{2} - (\alpha + 1) + \frac{\alpha \mathcal{L}}{2}. \quad (\text{ECL14})$$

Equations ECL6, ECL12, and ECL14 therefore evaluate the same Robin constant, logarithmic moment, and variational energy as the original EIQ integrals.

C Internal derivation of the exact Binet identity

This note proves the exact identity used in the actual boundary-tilt calculation from the defining Gamma integral, elementary integration, locally uniform products, and the Beta substitution. No Stirling formula, Gamma asymptotic, or cited Binet identity is assumed.

All complex powers of positive real integration variables use their real logarithms. The notation $\text{Log } z$ on $\Re z > 0$ denotes the principal holomorphic logarithm. The holomorphic logarithm of Gamma will be constructed below and will be real on the positive real axis.

C.1 1. Defining integrals and the finite Euler formula

For $\Re z > 0$, define

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx. \quad (\text{BI1})$$

This is absolutely convergent. On a compact set with $0 < a \leq \Re z \leq b$, its integrand and each of its first finitely many derivatives in z are bounded in absolute value by

$$(x^{a-1} + x^{b-1})(1 + |\log x|^j)e^{-x},$$

with the appropriate fixed integer j . This function is integrable at zero and infinity. Differentiation under the integral is therefore valid, and Γ is holomorphic on the right half-plane. Integration by parts gives

$$\Gamma(z+1) = z\Gamma(z), \quad \Gamma(1) = 1, \quad (\text{BI2})$$

because $x^z e^{-x}$ tends to zero at both endpoints for $\Re z > 0$.

For $\Re z, \Re w > 0$, let

$$B(z, w) = \int_0^1 t^{z-1} (1-t)^{w-1} dt. \quad (\text{BI3})$$

For an integer $N \geq 1$, integration by parts gives

$$B(z, N+1) = \frac{N}{z} B(z+1, N).$$

Iterating, and using $B(z+N, 1) = 1/(z+N)$, proves the exact finite formula

$$B(z, N+1) = \frac{N!}{z(z+1) \cdots (z+N)}. \quad (\text{BI4})$$

In particular none of these denominators vanishes in the right half-plane.

Set

$$E_N(z) = N^z B(z, N+1) = \frac{N^z N!}{z(z+1) \cdots (z+N)}, \quad N \geq 1. \quad (\text{BI5})$$

The substitution $t = x/N$ gives

$$E_N(z) = \int_0^N x^{z-1} (1-x/N)^N dx. \quad (\text{BI6})$$

For $0 \leq x < N$, $\log(1-x/N) \leq -x/N$, so

$$0 \leq (1-x/N)^N \leq e^{-x}.$$

For each fixed $x > 0$, the expression on the left tends to e^{-x} , with the indicator of $x < N$ understood.

On a compact set as above, therefore

$$\begin{aligned} \sup_z |E_N(z) - \Gamma(z)| &\leq \int_0^\infty (x^{a-1} + x^{b-1}) \\ &\quad \cdot |(1-x/N)^N \mathbf{1}_{\{x < N\}} - e^{-x}| dx \longrightarrow 0 \end{aligned}$$

by dominated convergence. The same argument with an additional factor $|\log x|^j$, for $j = 1, 2$, proves locally uniform convergence of those two derivatives as well. Thus the Euler limit and the derivative limits are established directly from their integrals.

C.2 2. Nonvanishing and the holomorphic Gamma logarithm

Write $H_N = \sum_{j=1}^N 1/j$. The real numbers

$$\gamma_N = H_N - \log N$$

decrease and are bounded below by zero. Indeed

$$\gamma_{N+1} - \gamma_N = \frac{1}{N+1} - \log(1 + 1/N) < 0$$

by integrating $1/x > 1/(N+1)$ on $[N, N+1)$, while $H_N \geq \int_1^{N+1} dx/x = \log(N+1)$. Let $\gamma_E = \lim_N \gamma_N$.

For $\Re z > 0$, every $1 + z/j$ has positive real part. Its principal logarithm is holomorphic. Define

$$Q_N(z) = \text{Log } z + z(H_N - \log N) + \sum_{j=1}^N (\text{Log}(1 + z/j) - z/j). \quad (\text{BI7})$$

Exponentiating this literal sum and using (BI5) gives exactly

$$e^{Q_N(z)} = zN^{-z} \prod_{j=1}^N (1 + z/j) = \frac{1}{E_N(z)}. \quad (\text{BI8})$$

The logarithmic series in (BI7) converges locally uniformly. To prove this explicitly, for $|w| \leq 1/2$,

$$\text{Log}(1 + w) - w = - \int_0^1 \frac{tw^2}{1 + tw} dt,$$

whence $|\text{Log}(1 + w) - w| \leq |w|^2$. On a compact set with $|z| \leq R$, this bounds the j -th summand by R^2/j^2 whenever $j \geq 2R$. The finite initial part is holomorphic and the summable bound proves uniform convergence of the tail. Hence

$$Q(z) = \text{Log } z + \gamma_E z + \sum_{j=1}^{\infty} (\text{Log}(1 + z/j) - z/j) \quad (\text{BI9})$$

is holomorphic and $Q_N \rightarrow Q$ locally uniformly.

Taking limits in $E_N e^{Q_N} = 1$, using Section 1, proves

$$\Gamma(z) e^{Q(z)} = 1. \quad (\text{BI10})$$

This proves that Gamma is nowhere zero on the entire right half-plane. It also constructs the required holomorphic logarithm:

$$\mathcal{G}(z) := -Q(z), \quad e^{\mathcal{G}(z)} = \Gamma(z). \quad (\text{BI11})$$

For $x > 0$, every term in $Q(x)$ is real and the defining Gamma integral is positive. Thus $\mathcal{G}(x)$ is precisely the real logarithm of $\Gamma(x)$; there is no unspecified additive $2\pi i$ constant.

The second derivatives of the series in (BI9) converge locally uniformly, since $\sum_j |z + j|^{-2}$ does. Termwise differentiation gives

$$\boxed{\mathcal{G}''(z) = \sum_{j=0}^{\infty} \frac{1}{(z + j)^2} \quad (\Re z > 0).} \quad (\text{BI12})$$

For justification without assuming a derivative-series theorem, on any slightly larger compact subset of the same half-plane the series converges uniformly; Cauchy's integral formula on small circles then gives uniform convergence of its derivatives. Direct estimates of the derivatives of its summands give the same conclusion.

C.3 3. Beta product, duplication, and the factor $\sqrt{\pi}$

The absolutely convergent product of the two defining Gamma integrals admits the change of variables

$$x = rt, \quad y = r(1 - t), \quad r > 0, \quad 0 < t < 1.$$

Its Jacobian matrix has determinant $-r$, so the positive integration Jacobian is r . Consequently

$$\Gamma(z)\Gamma(w) = \Gamma(z + w)B(z, w) \quad (\Re z, \Re w > 0). \quad (\text{BI13})$$

All powers here are powers of positive real bases. Fubini and the change of variables are justified by the same integrals with real parts of the exponents.

In $B(z, z)$, first substitute $t = (1 + u)/2$, and then use symmetry and $v = u^2$. Every factor is retained:

$$\begin{aligned} B(z, z) &= 2^{1-2z} \int_{-1}^1 (1 - u^2)^{z-1} du \\ &= 2^{2-2z} \int_0^1 (1 - u^2)^{z-1} du \\ &= 2^{1-2z} B(1/2, z). \end{aligned} \quad (\text{BI14})$$

Finally $x = t^2$ in (BI1) gives $\Gamma(1/2) = 2 \int_0^\infty e^{-t^2} dt$. The latter is the positive integral over the whole real line. Its square is $\int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy$. Polar coordinates, whose positive Jacobian is r , evaluate this square as

$$2\pi \int_0^\infty r e^{-r^2} dr = \pi.$$

Thus $\Gamma(1/2) = \sqrt{\pi}$. Inserting (BI13) into (BI14), and using the nonvanishing established in (BI10), proves

$$\boxed{\Gamma(z)\Gamma(z + 1/2) = 2^{1-2z} \sqrt{\pi} \Gamma(2z) \quad (\Re z > 0).} \quad (\text{BI15})$$

On the positive real axis all these factors are positive, so their real logarithms obey the corresponding additive identity without any branch ambiguity.

C.4 4. The bounded integral remainder

For $u > 0$, define

$$K(u) = \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{1}{u}. \quad (\text{BI16})$$

We prove its exact bounds rather than assuming a Bernoulli expansion. With $x = u/2 > 0$,

$$K(u) = \frac{\coth x - 1/x}{4x}.$$

The two functions

$$F_0(x) = x \cosh x - \sinh x, \quad F_1(x) = (x^2 + 3) \sinh x - 3x \cosh x$$

vanish at zero and satisfy $F_0'(x) = x \sinh x \geq 0$ and $F_1'(x) = xF_0(x) \geq 0$. Dividing their nonnegative values by positive factors gives

$$0 \leq \coth x - \frac{1}{x} \leq \frac{x}{3}.$$

Therefore

$$\boxed{0 \leq K(u) \leq \frac{1}{12} \quad (u > 0).} \quad (\text{BI17})$$

Define the actual remainder candidate

$$R(z) = \int_0^\infty e^{-zu} K(u) du, \quad \Re z > 0. \quad (\text{BI18})$$

It is absolutely convergent and holomorphic. Differentiating j times is justified on every compact set with $\Re z \geq a > 0$ by $u^j e^{-au}/12$. In particular

$$|R(z)| \leq \frac{1}{12\Re z}, \quad 0 \leq R(x) \leq \frac{1}{12x} \quad (x > 0). \quad (\text{BI19})$$

C.5 5. Equality of second derivatives

Put $c = \frac{1}{2} \log(2\pi)$, and define

$$\mathcal{F}(z) = (z - \frac{1}{2}) \text{Log } z - z + c + R(z). \quad (\text{BI20})$$

Differentiation and the exact algebraic identity

$$u^2 K(u) = \frac{u}{e^u - 1} - 1 + \frac{u}{2}$$

give

$$\begin{aligned} \mathcal{F}''(z) &= \frac{1}{z} + \frac{1}{2z^2} + \int_0^\infty e^{-zu} \left(\frac{u}{e^u - 1} - 1 + \frac{u}{2} \right) du \\ &= \frac{1}{z^2} + \int_0^\infty \frac{ue^{-zu}}{e^u - 1} du. \end{aligned} \quad (\text{BI21})$$

Here $\int_0^\infty e^{-zu} du = 1/z$ and $\int_0^\infty ue^{-zu} du = 1/z^2$ follow by elementary integration on $\Re z > 0$; all cancellation factors are displayed.

The geometric series $1/(e^u - 1) = \sum_{j=1}^\infty e^{-ju}$ is valid for $u > 0$. Its absolute integral after multiplication by ue^{-zu} is bounded by the same expression with $\Re z$, which is integrable at zero and infinity. Hence termwise integration is valid, and

$$\mathcal{F}''(z) = \frac{1}{z^2} + \sum_{j=1}^\infty \int_0^\infty ue^{-(z+j)u} du = \sum_{j=0}^\infty \frac{1}{(z+j)^2} = \mathcal{G}''(z). \quad (\text{BI22})$$

The right half-plane is connected, so the holomorphic function $\mathcal{F} - \mathcal{G}$ has zero second derivative and consequently equals $az + b$ there, for constants $a, b \in \mathbb{C}$.

C.6 6. The recurrence fixes the linear constant

For positive real x , (BI2) and the real-axis branch in (BI11) give

$$\mathcal{G}(x+1) - \mathcal{G}(x) = \log x.$$

The corresponding exact difference of the explicit function (BI20) is

$$\mathcal{F}(x+1) - \mathcal{F}(x) - \log x = (x + \frac{1}{2}) \log(1 + 1/x) - 1 + R(x+1) - R(x). \quad (\text{BI23})$$

The left side equals a . The elementary integral bounds

$$\frac{y}{1+y} \leq \log(1+y) \leq y \quad (y > 0)$$

show that $x \log(1 + 1/x) \rightarrow 1$ and $\frac{1}{2} \log(1 + 1/x) \rightarrow 0$. The remainder bound (BI19) makes the remaining difference tend to zero. Thus $a = 0$, and $\mathcal{F} - \mathcal{G} = b$ throughout the right half-plane.

C.7 7. Duplication fixes the additive constant

For $x > 0$, the positive-real logarithm of (BI15) is

$$\mathcal{G}(x) + \mathcal{G}(x + 1/2) - \mathcal{G}(2x) = (1 - 2x) \log 2 + \frac{1}{2} \log \pi. \quad (\text{BI24})$$

Since $\mathcal{F} = \mathcal{G} + b$, the difference between the same expression using $\mathcal{F}$ and the right side of (BI24) equals exactly $b + b - b = b$.

Insert (BI20) without dropping any constant. The terms containing $\log x$ cancel, and the resulting difference is

$$\begin{aligned} b &= x \log(1 + 1/(2x)) - \frac{1}{2} \\ &\quad + \left(c - \frac{1}{2} \log(2\pi)\right) + R(x) + R(x + 1/2) - R(2x). \end{aligned} \quad (\text{BI25})$$

Our displayed $c = \frac{1}{2} \log(2\pi)$ cancels exactly. The same elementary logarithm bounds show $x \log(1 + 1/(2x)) \rightarrow 1/2$, and each remainder tends to zero by (BI19). Thus $b = 0$.

Combining Sections 5–7 proves the complete exact identity

$$\boxed{\log \Gamma(z) = (z - \frac{1}{2}) \text{Log } z - z + \frac{1}{2} \log(2\pi) + \int_0^\infty e^{-zu} \left(\frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2} \right) \frac{du}{u}, \quad \Re z > 0.} \quad (\text{BI26})$$

The logarithm on the left is the uniquely specified holomorphic branch from (BI11). The integral is absolutely convergent, and its bound and all differentiations have been justified. This is the identity used by the actual boundary-tilt proof.